

5 **Draft Standard for**
6 **Local and metropolitan area networks—**

7 **Link Aggregation**

8 **Amendment 1:**
9 **YANG for Link Aggregation**

10 Prepared by the

11 **Time-Sensitive Networking (TSN) Task Group of IEEE 802.1**

12 Sponsor

13 **LAN/MAN Standards Committee**

14 **of the**

15 **IEEE Computer Society**

16 **This and the following cover pages are not part of the draft.** They provide revision and other information
17 for IEEE 802.1 Working Group members and participants in the IEEE Standards Association ballot process,
18 and will be updated as convenient.

19 The text proper of this draft begins with the [Title page](#).

Important Notice

This document is an unapproved draft of a proposed IEEE Standard. IEEE hereby grants the named IEEE SA Working Group or Standards Committee Chair permission to distribute this document to participants in the receiving IEEE SA Working Group or Standards Committee, for purposes of review for IEEE standardization activities. No further use, reproduction, or distribution of this document is permitted without the express written permission of IEEE Standards Association (IEEE SA). Prior to any review or use of this draft standard, in part or in whole, by another standards development organization, permission must first be obtained from IEEE SA (stds-copyright@ieee.org). This page is included as the cover of this draft, and shall not be modified or deleted.

IEEE Standards Association
445 Hoes Lane
Piscataway, NJ 08854, USA

1 Current draft

2 The text of this draft, following these cover pages, is that of P802.1AXdz, as updated by the resolution of
3 comments received during the initial SA Ballot on draft 2.0.

4 YANG modules

5 The YANG modules specified by this standard are attached to the draft pdf as plain text (UTF-8) .yang files.

6

7

Draft Standard for Local and metropolitan area networks— Link Aggregation Amendment 1: YANG for Link Aggregation

Prepared by the
Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

Sponsor
**LAN/MAN Standards Committee
of the
IEEE Computer Society**

Copyright © 2025 by the IEEE.
Three Park Avenue
New York, New York 10016-5997, USA

All rights reserved.

This document is an unapproved draft of a proposed IEEE Standard. As such, this document is subject to change. USE AT YOUR OWN RISK! IEEE copyright statements SHALL NOT BE REMOVED from draft or approved IEEE standards, or modified in any way. Because this is an unapproved draft, this document must not be utilized for any conformance/compliance purposes. Permission is hereby granted for officers from each IEEE Standards Working Group or Committee to reproduce the draft document developed by that Working Group for purposes of international standardization consideration. IEEE Standards Department must be informed of the submission for consideration prior to any reproduction for international standardization consideration (stds.ipr@ieee.org). Prior to adoption of this document, in whole or in part, by another standards development organization, permission must first be obtained from the IEEE Standards Department (stds.ipr@ieee.org). When requesting permission, IEEE Standards Department will require a copy of the standard development organization's document highlighting the use of IEEE content. Other entities seeking permission to reproduce this document, in whole or in part, must also obtain permission from the IEEE Standards Department.

IEEE Standards Department
445 Hoes Lane
Piscataway, NJ 08854, USA

1

2 **Abstract:** This amendment to IEEE Std 802.1AX-2020 specifies a Unified Modeling Language
3 (UML)-based model and YANG modules for Link Aggregation configuration and status reporting.

4 **Keywords:** Aggregated Link, Aggregator, Distributed Resilient Network Interconnect, DRNI,
5 interconnect, Link Aggregation, Link Aggregation Group, local area network, management,
6 Network-Network Interface, NNI, YANG.

7

The Institute of Electrical and Electronics Engineers, Inc.
3 Park Avenue, New York, NY 10016-5997, USA

Copyright © 2025 by the Institute of Electrical and Electronics Engineers, Inc.
All rights reserved. Unapproved draft.

OMG®, UML®, and Unified Modeling Language™ are either registered trademarks or trademarks of Object Management Group, Inc. in the United States and/or other countries.
IEEE and 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by the Institute of Electrical and Electronics Engineers, Incorporated.

PDF: ISBN 978-X-XXX-XXX-X STDXXXXX
Print: ISBN 978-X-XXX-XXX-X STDPDXXXXX

IEEE prohibits discrimination, harassment, and bullying.

For more information, visit <http://www.ieee.org/web/aboutus/whatis/policies/p9-26.html>.

No part of this publication may be reproduced in any form, in an electronic retrieval system or otherwise, without the prior written permission of the publisher.

1 Important Notices and Disclaimers Concerning IEEE Standards 2 Documents

3 IEEE Standards documents are made available for use subject to important notices and legal disclaimers.
4 These notices and disclaimers, or a reference to this page (<https://standards.ieee.org/ipr/disclaimers.html>),
5 appear in all standards and may be found under the heading “Important Notices and Disclaimers Concerning
6 IEEE Standards Documents.”

7 Notice and Disclaimer of Liability Concerning the Use of IEEE Standards 8 Documents

9 IEEE Standards documents are developed within IEEE Societies and subcommittees of IEEE Standards
10 Association (IEEE SA) Board of Governors. IEEE develops its standards through an accredited consensus
11 development process, which brings together volunteers representing varied viewpoints and interests to
12 achieve the final product. IEEE Standards are documents developed by volunteers with scientific, academic,
13 and industry-based expertise in technical working groups. Volunteers are not necessarily members of IEEE
14 or IEEE SA and participate without compensation from IEEE. While IEEE administers the process and
15 establishes rules to promote fairness in the consensus development process, IEEE does not independently
16 evaluate, test, or verify the accuracy of any of the information or the soundness of any judgments contained
17 in its standards.

18 IEEE makes no warranties or representations concerning its standards, and expressly disclaims all
19 warranties, express or implied, concerning this standard, including but not limited to the warranties of
20 merchantability, fitness for a particular purpose and non-infringement. In addition, IEEE does not warrant or
21 represent that the use of the material contained in its standards is free from patent infringement. IEEE
22 standards documents are supplied “AS IS” and “WITH ALL FAULTS.”

23 Use of an IEEE standard is wholly voluntary. The existence of an IEEE Standard does not imply that there
24 are no other ways to produce, test, measure, purchase, market, or provide other goods and services related to
25 the scope of the IEEE standard. Furthermore, the viewpoint expressed at the time a standard is approved and
26 issued is subject to change brought about through developments in the state of the art and comments
27 received from users of the standard.

28 In publishing and making its standards available, IEEE is not suggesting or rendering professional or other
29 services for, or on behalf of, any person or entity, nor is IEEE undertaking to perform any duty owed by any
30 other person or entity to another. Any person utilizing any IEEE Standards document, should rely upon his or
31 her own independent judgment in the exercise of reasonable care in any given circumstances or, as appropriate,
32 seek the advice of a competent professional in determining the appropriateness of a given IEEE standard.

33 IN NO EVENT SHALL IEEE BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL,
34 EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO: THE
35 NEED TO PROCURE SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR
36 BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY,
37 WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR
38 OTHERWISE) ARISING IN ANY WAY OUT OF THE PUBLICATION, USE OF, OR RELIANCE UPON
39 ANY STANDARD, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE AND
40 REGARDLESS OF WHETHER SUCH DAMAGE WAS FORESEEABLE.

41 Translations

42 The IEEE consensus development process involves the review of documents in English only. In the event
43 that an IEEE standard is translated, only the English version published by IEEE is the approved IEEE
44 standard.

1 Official statements

2 A statement, written or oral, that is not processed in accordance with the IEEE SA Standards Board
3 Operations Manual shall not be considered or inferred to be the official position of IEEE or any of its
4 committees and shall not be considered to be, nor be relied upon as, a formal position of IEEE. At lectures,
5 symposia, seminars, or educational courses, an individual presenting information on IEEE standards shall
6 make it clear that the presenter's views should be considered the personal views of that individual rather
7 than the formal position of IEEE, IEEE SA, the Standards Committee, or the Working Group. Statements
8 made by volunteers may not represent the formal position of their employer(s) or affiliation(s).

9 Comments on standards

10 Comments for revision of IEEE Standards documents are welcome from any interested party, regardless of
11 membership affiliation with IEEE or IEEE SA. However, **IEEE does not provide interpretations,**
12 **consulting information, or advice pertaining to IEEE Standards documents.**

13 Suggestions for changes in documents should be in the form of a proposed change of text, together with
14 appropriate supporting comments. Since IEEE standards represent a consensus of concerned interests, it is
15 important that any responses to comments and questions also receive the concurrence of a balance of interests.
16 For this reason, IEEE and the members of its Societies and subcommittees of the IEEE SA Board of
17 Governors are not able to provide an instant response to comments, or questions except in those cases where
18 the matter has previously been addressed. For the same reason, IEEE does not respond to interpretation
19 requests. Any person who would like to participate in evaluating comments or in revisions to an IEEE standard
20 is welcome to join the relevant IEEE working group. You can indicate interest in a working group using the
21 Interests tab in the Manage Profile & Interests area of the [IEEE SA myProject system](#).¹ An IEEE Account is
22 needed to access the application.

23 Comments on standards should be submitted using the [Contact Us](#) form.²

24 Laws and regulations

25 Users of IEEE Standards documents should consult all applicable laws and regulations. Compliance with the
26 provisions of any IEEE Standards document does not constitute compliance to any applicable regulatory
27 requirements. Implementers of the standard are responsible for observing or referring to the applicable
28 regulatory requirements. IEEE does not, by the publication of its standards, intend to urge action that is not
29 in compliance with applicable laws, and these documents may not be construed as doing so.

30 Data privacy

31 Users of IEEE Standards documents should evaluate the standards for considerations of data privacy and
32 data ownership in the context of assessing and using the standards in compliance with applicable laws and
33 regulations.

34 Copyrights

35 IEEE draft and approved standards are copyrighted by IEEE under US and international copyright laws.
36 They are made available by IEEE and are adopted for a wide variety of both public and private uses. These
37 include both use, by reference, in laws and regulations, and use in private self-regulation, standardization,
38 and the promotion of engineering practices and methods. By making these documents available for use and
39 adoption by public authorities and private users, neither IEEE nor its licensors waive any rights in copyright
40 to the documents.

¹ Available at: <https://development.standards.ieee.org/myproject-web/public/view.html#landing>.

² Available at: <https://standards.ieee.org/content/ieee-standards/en/about/contact/index.html>.

1 Photocopies

2 Subject to payment of the appropriate licensing fees, IEEE will grant users a limited, non-exclusive license
3 to photocopy portions of any individual standard for company or organizational internal use or individual,
4 non-commercial use only. To arrange for payment of licensing fees, please contact Copyright Clearance
5 Center, Customer Service, 222 Rosewood Drive, Danvers, MA 01923 USA; +1 978 750 8400;
6 <https://www.copyright.com/>. Permission to photocopy portions of any individual standard for educational
7 classroom use can also be obtained through the Copyright Clearance Center.

8 Updating of IEEE Standards documents

9 Users of IEEE Standards documents should be aware that these documents may be superseded at any time
10 by the issuance of new editions or may be amended from time to time through the issuance of amendments,
11 corrigenda, or errata. An official IEEE document at any point in time consists of the current edition of the
12 document together with any amendments, corrigenda, or errata then in effect.

13 Every IEEE standard is subjected to review at least every 10 years. When a document is more than 10 years
14 old and has not undergone a revision process, it is reasonable to conclude that its contents, although still of
15 some value, do not wholly reflect the present state of the art. Users are cautioned to check to determine that
16 they have the latest edition of any IEEE standard.

17 In order to determine whether a given document is the current edition and whether it has been amended
18 through the issuance of amendments, corrigenda, or errata, visit [IEEE Xplore](#) or [contact IEEE](#).³ For more
19 information about the IEEE SA or IEEE's standards development process, visit the IEEE SA Website.

20 Errata

21 Errata, if any, for all IEEE standards can be accessed on the [IEEE SA Website](#).⁴ Search for standard number
22 and year of approval to access the web page of the published standard. Errata links are located under the
23 Additional Resources Details section. Errata are also available in [IEEE Xplore](#). Users are encouraged to
24 periodically check for errata.

25 Patents

26 IEEE Standards are developed in compliance with the [IEEE SA Patent Policy](#).⁵

27 Attention is called to the possibility that implementation of this standard may require use of subject matter
28 covered by patent rights. By publication of this standard, no position is taken by the IEEE with respect to the
29 existence or validity of any patent rights in connection therewith. If a patent holder or patent applicant has
30 filed a statement of assurance via an Accepted Letter of Assurance, then the statement is listed on the
31 IEEE SA Website at <https://standards.ieee.org/about/sasb/patcom/patents.html>. Letters of Assurance may
32 indicate whether the Submitter is willing or unwilling to grant licenses under patent rights without
33 compensation or under reasonable rates, with reasonable terms and conditions that are demonstrably free of
34 any unfair discrimination to applicants desiring to obtain such licenses.

35 Essential Patent Claims may exist for which a Letter of Assurance has not been received. The IEEE is not
36 responsible for identifying Essential Patent Claims for which a license may be required, for conducting
37 inquiries into the legal validity or scope of Patents Claims, or determining whether any licensing terms or
38 conditions provided in connection with submission of a Letter of Assurance, if any, or in any licensing
39 agreements are reasonable or non-discriminatory. Users of this standard are expressly advised that
40 determination of the validity of any patent rights, and the risk of infringement of such rights, is entirely their
41 own responsibility. Further information may be obtained from the IEEE Standards Association.

³ Available at: <https://ieeexplore.ieee.org/browse/standards/collection/ieee>.

⁴ Available at: <https://standards.ieee.org/standard/index.html>.

⁵ Available at: <https://standards.ieee.org/about/sasb/patcom/materials.html>.

1 **IMPORTANT NOTICE**

2 IEEE Standards do not guarantee or ensure safety, security, health, or environmental protection, or ensure
3 against interference with or from other devices or networks. IEEE Standards development activities consider
4 research and information presented to the standards development group in developing any safety
5 recommendations. Other information about safety practices, changes in technology or technology
6 implementation, or impact by peripheral systems also may be pertinent to safety considerations during
7 implementation of the standard. Implementers and users of IEEE Standards documents are responsible for
8 determining and complying with all appropriate safety, security, environmental, health, and interference
9 protection practices and all applicable laws and regulations.

1 Participants

2 <<The following lists will be updated in the usual way prior to publication>>

3 At the time this standard was submitted to the IEEE-SA Standards Board for approval, the IEEE 802.1
4 Working Group had the following membership:

5 **Glenn Parsons**, *Chair*
6 **Jessy V. Rouyer**, *Vice Chair*
7 **János Farkas**, *Chair, Time-Sensitive Networking Task Group*
8 **David McCall**, *Vice Chair, Time-Sensitive Networking Task Group*
9 **Stephen Haddock**, *Editor*
10

<<TBA>>

¹ The following members of the individual balloting committee voted on this standard. Balloters may have
² voted for approval, disapproval, or abstention.

<<TBA>>

³ When the IEEE-SA Standards Board approved this standard on XX Month 20xx, it had the following
⁴ membership:

⁵ <<TBA>>

<<TBA>>

⁶
⁷ *Member Emeritus
⁸
⁹
¹⁰

1 Introduction

This introduction is not part of IEEE Std 802.1AXdz™-20XX, IEEE Standard for Local and metropolitan area networks—Link Aggregation—Amendment 1: YANG for Link Aggregation.

2 IEEE Std 802.1AXdz™-202X: YANG for Link Aggregation specifies a Unified Modeling Language
3 (UML)-based model and YANG modules for Link Aggregation configuration and status reporting

4 This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution.
5 Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and
6 to incorporate new related material. Information on the current revision state of this and other IEEE 802
7 standards may be obtained from

8 Secretary, IEEE-SA Standards Board
9 445 Hoes Lane
10 Piscataway, NJ 08854-4141
11 USA

Contents

2.	Normative references	15
4.	Acronyms and abbreviations	16
5.	Conformance.....	17
5.3.2	Link Aggregation options	17
10.	YANG Data Model.....	18
10.1	YANG Framework	18
10.2	Information Model for Link Aggregation Management.....	18
10.2.1	Link Aggregation model.....	19
10.2.2	DRNI model.....	22
10.3	Structure of the Link Aggregation YANG data model.....	23
10.4	Relationship to other YANG modules.....	23
10.4.1	IEEE 802.1AX Types Module.....	23
10.4.2	IETF YANG Types Module	23
10.4.3	IETF Interfaces YANG Module	23
10.4.4	IEEE 802 Types Module.....	23
10.5	Security Considerations	24
10.5.1	Security considerations of the ieee802-dot1ax YANG modules	24
10.6	YANG schema tree definitions.....	24
10.6.1	Schema for the ieee802-dot1ax-linkagg YANG module.....	25
10.6.2	Schema for the ieee802-dot1ax-drni YANG module	26
10.7	YANG modules,,	27
10.7.1	The ieee802-dot1ax-types YANG module	27
10.7.2	The ieee802-dot1ax-linkagg YANG module.....	34
10.7.3	The ieee802-dot1ax-drni YANG module	47
Annex A		57
(normative)		57
Protocol implementation conformance statement (PICS) proforma.....		57
A.2.1	Major capabilities/options	57
Annex G		58
(informative)		58
Bibliography		58

Figures

Figure 10-1	YANG root hierarchy with Link Aggregation YANG modules.....	19
Figure 10-2	Link Aggregation Configuration and Monitoring Objects	21
Figure 10-3	DRNI Configuration and Monitoring Objects	22

Tables

Table 10-1	Structure of the YANG modules	23
------------	-------------------------------------	----

¹ 2. Normative references

² *Insert the following references into Clause 2 in alphanumeric order:*

³ IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011.

⁴ IETF RFC 6991, Common YANG Data Types, July 2013

⁵ IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

⁶ IETF RFC 8343, A YANG Data Model for Interface Management, March 2018

⁷

¹ 4. Acronyms and abbreviations

² *Insert the following acronyms into Clause 4 in alphabetical order:*

³ NETCONF Network Configuration Protocol

⁴ UML® Unified Modeling Language™¹

¹ UML® is a registered trademark of the Object Management Group, Inc., and Unified Modeling Language™ is a trademark of the Object Management Group, Inc.

1 5. Conformance

2 *Insert the following text (items j and k) after item i) in the lettered list of 5.3.2*

3 5.3.2 Link Aggregation options

- 4 j) Support YANG modules for the management of Link Aggregation capabilities (Clause 10).
- 5 k) Support YANG modules for the management of DRNI capabilities (Clause 10).

1 *Insert Clause 10 after Clause 9 as follows:*

2 **10. YANG Data Model**

3 This clause specifies the YANG (IETF RFC 7950) data model that provides control and status monitoring of
4 systems and system components that implement functionality specified in this standard.

5 This clause:

- 6 a) Introduces the YANG framework that governs the naming and hierarchy of configuration and
7 operational data structures in the data models, and the modeling of network interfaces (10.1).
- 8 b) Describes the information data model and its relationship to the operational processes and managed
9 objects specified in the other clauses of this standard, and provides a representation, similar to the
10 Unified Modeling Language (UML), of relevant objects (10.2).
- 11 c) Describes the structure of the YANG data model, which comprises three YANG modules (10.3).
- 12 d) Includes a relationship description of other modules imported in YANG modules (10.4)
- 13 e) Reviews security considerations applicable to each of the modules, with specific reference to data
14 nodes in the YANG modules that compose the YANG data model (10.5).
- 15 f) Includes each of the YANG modules (10.7) and its data schema (10.6).

16 **10.1 YANG Framework**

17 The YANG framework applies hierarchy in the following areas:

- 18 a) The uniform resource name (URN), as specified in IEEE Std 802 for the IEEE 802 series of
19 standards.
- 20 b) The YANG objects form a hierarchy of configuration and operational data structures that specify the
21 YANG data model.

22 **10.2 Information Model for Link Aggregation Management**

23 The YANG objects are based on the managed objects in Clause 7. A UML-like representation of the
24 management model is provided in the following subclauses.

25 NOTE—OMG® UML2.5² [B11] conventions together with C++ language constructs are used in this clause as a
26 representation to convey model structure and relationships..

27 The purpose of an UML-like³ diagram is to express the model design on a single piece of paper. The
28 structure of the UML-like representation shows the name of the object followed by a list of properties for the
29 object. The properties indicate its type and accessibility. The UML-like representation is meant to express
30 simplified semantics for the properties. It is not meant to provide the specific datatype as used to encode the
31 object in either MIB or YANG. In the UML-like representation, a box with a white background represents
32 information that comes from sources outside of the IEEE. A box with a gray background represents objects
33 that are specified by this IEEE Standard.

34 The YANG hierarchical structure that incorporates the Link Aggregation YANG modules supported by this
35 standard is represented by Figure 10-1. In the figures in this clause, items that are shaded gray are described

²OMG® is a registered trademark of the Object Management Group, Inc.

³A description of the UML-like diagrams used in this clause is provided at <https://1.ieee802.org/uml-like-diagrams>.

1 in this document, items with no background shading are specified elsewhere. The YANG data model is
2 realized in three YANG modules. One module *ieee802-dot1ax-types* provides data types that are needed by
3 the Link Aggregation configuration and monitoring objects. The *ieee802-dot1ax-linkagg* module provides
4 the Link Aggregation, Aggregation Port, and Aggregator configuration and monitoring objects. The
5 *ieee802-dot1ax-drni* module provides the Distributed Resilient Network Interface (DRNI) configuration and
6 monitoring objects. The Link Aggregation and DRNI capabilities are not only applicable to IEEE Std
7 802.1Q bridges, but also (for example) end stations, and routers.

8

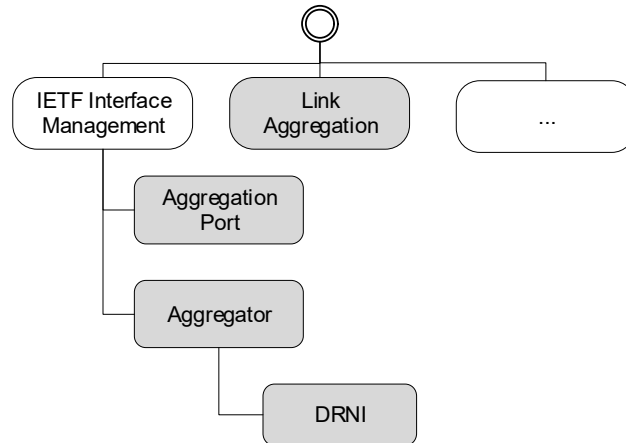


Figure 10-1—YANG root hierarchy with Link Aggregation YANG modules

9 10.2.1 Link Aggregation model

10 The Link Aggregation Configuration and Monitoring Objects in Figure 10-2 show the objects that are
11 applicable on a system supporting link aggregation, and the interfaces supporting link aggregation.

12 The objects on the system consist of a list of Aggregation Systems, and a list of key groups. A key group is
13 the set of Aggregators and Aggregation Ports that share the same System Priority, System MAC Address,
14 and Aggregation Key, and therefore can potentially form a Link Aggregation Group. Each key group has a
15 unique combination of actor-admin-key and actor-system, and includes parameters that have the same value
16 for any Aggregator and/or Aggregation Port that have the same actor-admin-key and actor-system.

17 NOTE—See 6.3.5 and 6.3.6 for a description of the identification of Aggregation Ports that can potentially form a Link
18 Aggregation Group, and Figure 6-15 for examples.

19 Each Aggregation Port is an interface (e.g. of type ethernetCsmacd) augmented with an “aggport” presence
20 container as shown on the right side of Figure 10-2. Each Aggregator is an interface to the entire Link
21 Aggregation Group (using type *ieee8023adlag*⁴) augmented with a “lag” container as shown on the left side
22 of Figure 10-2. The binding of Aggregation Ports to Aggregators is a dynamic function of the Link
23 Aggregation Control Protocol. The Aggregation Ports attached to an Aggregator can be read from the *lower-*
24 *layer-if* attribute of the Aggregator interface. The Aggregator to which an Aggregation Port is attached can
25 be read through the *higher-layer-if* attribute of the Aggregation Port interface. An Aggregation Port only
26 transmits and responds to LACPDUs when the interface “enabled” leaf is true, and only attaches to an
27 Aggregator when the Aggregator interface “enabled” leaf is true. The interface “oper-status” leaf of an
28 Aggregation Port is “up” when the Aggregation Port is attached to an Aggregator and the “collecting” and

⁴This type reflects the fact that the Link Aggregation specification was originally standardized in IEEE Std 802.3ad-2000. Further development has taken place in the 802.1 Working Group since 2008. Subsequent revisions have been published as IEEE Std 802.1AX.

¹ "distributing" bits of the "actor-oper-state" leaf are both true. The interface "oper-status" leaf of an
² Aggregator is "up" when it has at least one attached Aggregation Port that is "up".

1

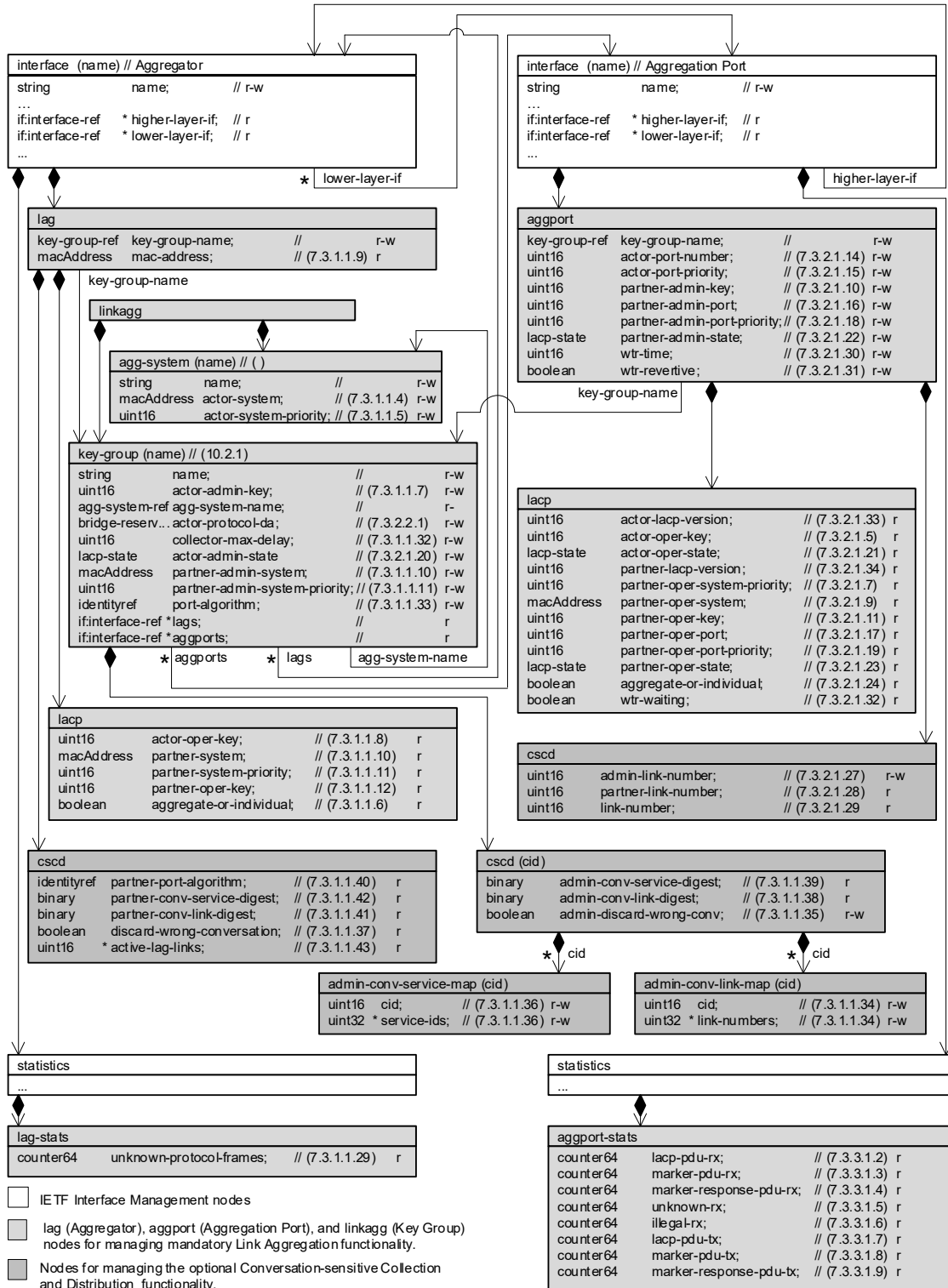


Figure 10-2—Link Aggregation Configuration and Monitoring Objects

1 10.2.2 DRNI model

2 The DRNI Configuration and Monitoring Objects in Figure 10-3 show the objects that are applicable to an
3 Aggregator augmented with a “drni” container.

4

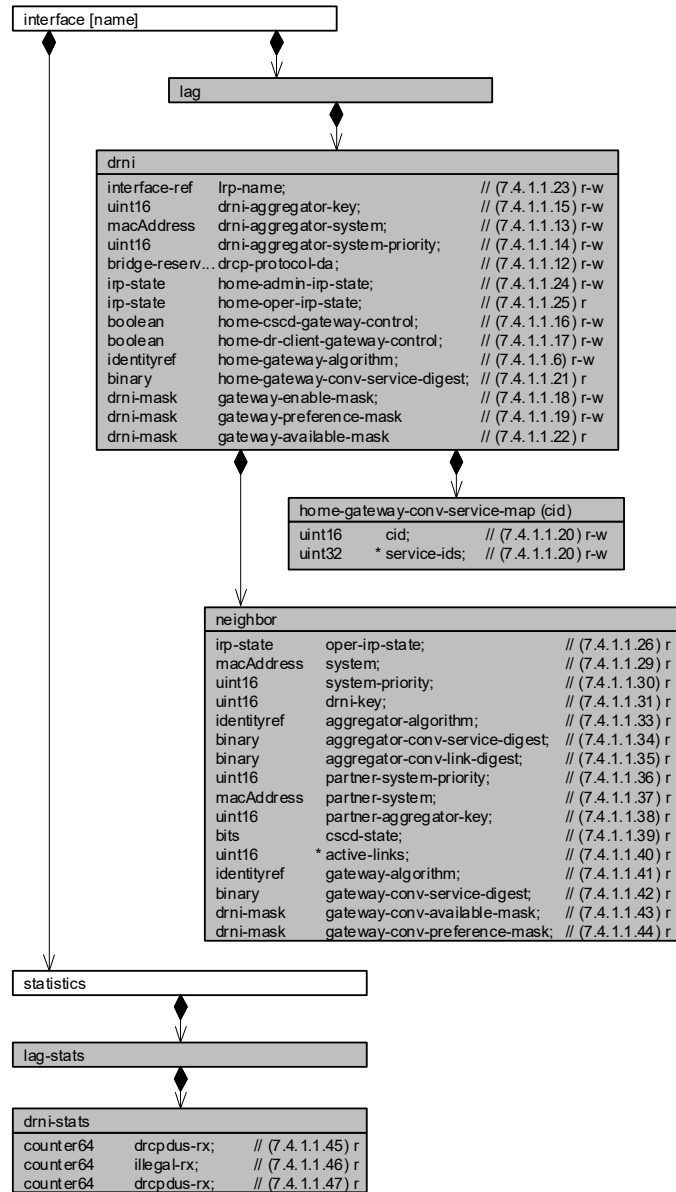


Figure 10-3—DRNI Configuration and Monitoring Objects

10.3 Structure of the Link Aggregation YANG data model

The YANG data model specified in this standard is comprised of three YANG modules. A summary of the modules contained in this clause is represented in Table 10-1.

Table 10-1—Structure of the YANG modules

Module	Subclause	Notes
ieee802-dot1ax-types	10.7.1	Type definitions used for Link Aggregation YANG.
ieee802-dot1ax-linkagg	10.7.2	Link Aggregation Management
ieee802-dot1ax-drni	10.7.3	DRNI Management

In the YANG modules below, if any discrepancy between the DESCRIPTION text and the corresponding specification in any other part of this standard occurs, the specifications outside this clause take precedence.

10.4 Relationship to other YANG modules

This clause describes how the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules are related to the YANG modules that are imported.

10.4.1 IEEE 802.1AX Types Module

The *ieee802-dot1ax-types* module provides reusable types that are used by the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* modules.

10.4.2 IETF YANG Types Module

The *ietf-yang-types* YANG module (IETF RFC 6991) contains a set of derived YANG types. This document leverages counter64.

10.4.3 IETF Interfaces YANG Module

The *ietf-interfaces* YANG module (IETF RFC 8343) contains a set of YANG definitions for managing network interfaces. This document augments an *ietf-interfaces:interface* with aggregation-port or aggregator data nodes..

10.4.4 IEEE 802 Types Module

The *ieee802-types* module provides reusable types that are used in IEEE 802 standards.

NOTE—The type for mac-addresses defined in *ieee802-types* has a pattern that allows upper and lower case letters. To avoid issues with string comparison, it is suggested to only use upper case for the letters in the hexadecimal numbers. Implementers using code comparing MAC addresses are advised that there is still an issue with a difference between the IETF mac-address definition and the IEEE mac-address definition.

10.5 Security Considerations

The YANG modules specified in this clause are designed to be accessed via a network configuration protocol (e.g., NETCONF protocol). In the case of NETCONF, the lowest NETCONF layer is the secure transport layer and the mandatory to implement secure transport is SSH. The NETCONF access control model provides the means to restrict access for particular NETCONF users to a pre-configured subset of all available NETCONF protocol operations and content.

It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those users who have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

10.5.1 Security considerations of the *ieee802-dot1ax* YANG modules

There are several management objects specified in the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules that are configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects could be considered sensitive or vulnerable in some network environments. A network configuration protocol, such as NETCONF (IETF RFC 6241), can support protocol operations that can edit or delete YANG module configuration data (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure environment without proper protection, then negative effects on the network operation are possible.

The following containers, and the objects in these containers, of the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules can be manipulated to interfere with the operation of the Link Aggregation Control Protocol (LACP) or the Distributed Relay Control Protocol (DRCP). This could, for example, cause network instability and result in the loss of service for a large number of end users.

```
dot1ax-linkagg/linkagg/key-group
dot1ax-linkagg/linkagg/key-group/cscd
dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-service-map
dot1ax-linkagg/linkagg/key-group/cscd/admin-conv-link-map
dot1ax-linkagg/aggport
dot1ax-linkagg/aggport/cscd/admin-link-number
dot1ax-linkagg/lag
dot1ax-drni/lag/drni
dot1ax-drni/lag/drni/home-gateway-conv-service-map
```

Some of the readable data in this YANG module could be considered sensitive or vulnerable in some network environments. It is important to control all types of access (e.g., including NETCONF get, get-config operations) to these objects and possibly to even encrypt the values of these objects when sending them over the network. For example, the system name and other information about the remote systems could provide information about the configuration and topology of the network and could be considered a privacy threat.

10.6 YANG schema tree definitions

A simplified graphical representation of the data model is used in this document. The meaning of the symbols in these diagrams is as follows:

- Brackets "[" and "]" enclose list keys.
- Abbreviations before data node names: "rw" means configuration (read-write), and "ro" means state data (read-only).

- 1 — Symbols after data node names: "?" means an optional node, "!" means a presence container, and "*" denotes a list and leaf-list.
- 2
- 3 — Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (":").
- 4 — Ellipsis ("...") stand for contents of subtrees that are not shown.

5 10.6.1 Schema for the *ieee802-dot1ax-linkagg* YANG module

```

6 module: ieee802-dot1ax-linkagg
7   +--rw linkagg
8     +--rw agg-system* [name]
9       | +--rw name string
10      | +--rw actor-system? ieee:mac-address
11      | +--rw actor-system-priority? uint16
12      +--rw key-group* [name]
13        +--rw name string
14        +--rw actor-admin-key uint16
15        +--rw agg-system-name agg-system-ref
16        +--rw actor-protocol-da? ax:bridge-reserved-addresses
17        +--rw collector-max-delay? uint16
18        +--rw actor-admin-state? ax:lacp-state
19        +--rw partner-admin-system? ieee:mac-address
20        +--rw partner-admin-system-priority? uint16
21        +--rw port-algorithm? identityref
22        +--ro lags* if:interface-ref
23        +--ro aggports* if:interface-ref
24        +--rw cscd {ax:cscd}?
25          +--rw admin-conv-service-map {ax:sid-map}?
26            | +--rw (method)?
27              | +--:(pattern)
28              | | +--rw pattern? identityref
29              | +--:(cid-list)
30              | | +--rw cid-list* [cid]
31              | | +--rw cid uint16
32              | | +--rw service-ids* uint32
33              +--ro admin-conv-service-digest? binary
34              +--rw admin-conv-link-map
35                | +--rw (method)?
36                | +--:(pattern)
37                | | +--rw pattern? identityref
38                | +--:(cid-list)
39                | | +--rw cid-list* [cid]
40                | | +--rw cid uint16
41                | | +--rw link-numbers* uint16
42                +--ro admin-conv-link-digest? binary
43                +--rw admin-discard-wrong-conv? enumeration {ax:dwc}?
44
45      augment /if:interfaces/if:interface:
46        +--rw lag
47          +--rw key-group-name key-group-ref
48          +--ro mac-address? ieee:mac-address
49          +--ro lacp
50            | +--ro actor-oper-key? uint16
51            | +--ro partner-system? ieee:mac-address
52            | +--ro partner-system-priority? uint16
53            | +--ro partner-oper-key? uint16
54            | +--ro aggregate-or-individual? boolean
55            +--ro cscd {ax:cscd}?
56              +--ro partner-port-algorithm? identityref
57              +--ro partner-conv-service-digest? binary
58              +--ro partner-conv-link-digest? binary
59              +--ro discard-wrong-conversation? boolean
60              +--ro active-lag-links* uint16
61      augment /if:interfaces/if:interface/if:statistics:
62        +--ro lag-stats
63          +--ro unknown-protocol-frames? yang:counter64
64      augment /if:interfaces/if:interface:
65        +--rw aggport!
66          +--rw key-group-name key-group-ref
67          +--rw actor-port-number? uint16

```

```

1      +---rw actor-port-priority?          uint16
2      +---rw partner-admin-key?            uint16
3      +---rw partner-admin-port?          uint16
4      +---rw partner-admin-port-priority?  uint16
5      +---rw partner-admin-state?         ax:lacp-state
6      +---rw wtr-time?                    uint16
7      +---rw wtr-revertive?               boolean
8      +---ro lacp
9      |   +---ro actor-lacp-version?      uint8
10     |   +---ro actor-oper-key?          uint16
11     |   +---ro actor-oper-state?        ax:lacp-state
12     |   +---ro partner-lacp-version?    uint8
13     |   +---ro partner-oper-system-priority?  uint16
14     |   +---ro partner-oper-system?     ieee:mac-address
15     |   +---ro partner-oper-key?        uint16
16     |   +---ro partner-oper-port?       uint16
17     |   +---ro partner-oper-port-priority?  uint16
18     |   +---ro partner-oper-state?      ax:lacp-state
19     |   +---ro aggregate-or-individual?  boolean
20     |   +---ro wtr-waiting?             boolean
21     +---rw cscd {ax:cscd}?
22         +---rw admin-link-number?      uint16
23         +---ro partner-link-number?     uint16
24         +---ro link-number?            uint16
25     augment /if:interfaces/if:interface/if:statistics:
26         +---ro aggport-stats
27             +---ro lacp-pdu-rx?          yang:counter64
28             +---ro marker-pdu-rx?        yang:counter64
29             +---ro marker-response-pdu-rx?  yang:counter64
30             +---ro unknown-rx?           yang:counter64
31             +---ro illegal-rx?           yang:counter64
32             +---ro lacp-pdu-tx?          yang:counter64
33             +---ro marker-pdu-tx?        yang:counter64
34             +---ro marker-response-pdu-tx?  yang:counter64
35

```

10.6.2 Schema for the *ieee802-dot1ax-drni* YANG module

```

37 module: ieee802-dot1ax-drni
38
39     augment /if:interfaces/if:interface/dot1ax:lag:
40         +---rw drni!
41             +---rw irp-name                if:interface-ref
42             +---rw drni-aggregator-key?    uint16
43             +---rw drni-aggregator-system?  ieee:mac-address
44             +---rw drni-aggregator-system-priority?  uint16
45             +---rw drcp-protocol-da?       ax:bridge-reserved-addresses
46             +---rw home-admin-irp-state?   irp-state
47             +---ro home-oper-irp-state?    irp-state
48             +---rw home-cscd-gateway-control?  boolean
49             +---rw home-dr-client-gateway-control?  boolean
50             +---rw home-gateway-algorithm?    identityref
51             +---rw home-gateway-conv-service-map {ax:sid-map}?
52             |   +---rw (method)?
53             |   |   +---:(pattern)
54             |   |   |   +---rw pattern?    identityref
55             |   |   +---:(cid-list)
56             |   |   |   +---rw cid-list* [cid]
57             |   |   |   |   +---rw cid      uint16
58             |   |   |   |   +---rw service-ids*  uint32
59             +---ro home-gateway-conv-service-digest?  binary
60             +---rw gateway-enable-mask
61             |   +---rw cid-str*            string
62             |   +---rw invert-mask?        boolean
63             +---rw gateway-preference-mask
64             |   +---rw cid-str*            string
65             |   +---rw invert-mask?        boolean
66             +---ro gateway-available-mask
67             |   +---ro cid-str*            string
68             |   +---ro invert-mask?        boolean
69             +---ro neighbor

```

```

1      +--ro oper-irp-state?          irp-state
2      +--ro system?                 ieee:mac-address
3      +--ro system-priority?         uint16
4      +--ro drni-key?                uint16
5      +--ro aggregator-algorithm?    identityref
6      +--ro aggregator-conv-service-digest? binary
7      +--ro aggregator-conv-link-digest? binary
8      +--ro partner-system-priority? uint16
9      +--ro partner-system?          ieee:mac-address
10     +--ro partner-aggregator-key?   uint16
11     +--ro cscd-state?              bits
12     +--ro active-links*             uint16
13     +--ro gateway-algorithm?        identityref
14     +--ro gateway-conv-service-digest? binary
15     +--ro gateway-available-mask
16     | +--ro cid-str*                string
17     | +--ro invert-mask?            boolean
18     +--ro gateway-preference-mask
19     | +--ro cid-str*                string
20     | +--ro invert-mask?            boolean
21     augment /if:interfaces/if:interface/if:statistics/dotlax:lag-stats:
22     +--ro drni-stats
23     | +--ro drcpdus-rx?             yang:counter64
24     | +--ro illegal-rx?             yang:counter64
25     | +--ro drcpdus-tx?             yang:counter64
26

```

27 10.7 YANG modules^{5,6,7}

28 10.7.1 The *ieee802-dot1ax-types* YANG module

```

29 module ieee802-dot1ax-types {
30   yang-version 1.1;
31   namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-types";
32   prefix dot1ax-types;
33
34   organization
35     "IEEE 802.1 Working Group";
36   contact
37     "WG-URL: http://www.ieee802.org/1/
38     WG-Email: stds-802-1-1@ieee.org
39
40     Contact: IEEE 802.1 Working Group Chair
41     Postal: C/O IEEE 802.1 Working Group
42             IEEE Standards Association
43             45 Hoes Lane
44             Piscataway, NJ 08854
45             USA
46     E-mail: stds-802-1-chairs@ieee.org";
47   description
48     "Common types used within 802.1AX Link Aggregation modules.
49
50     Copyright (C) IEEE (2025).
51
52     This version of this YANG module is part of IEEE Std 802.1AX;
53     see the standard itself for full legal notices.";
54
55   revision 2025-05-22 {
56     description
57       "Published as part of IEEE Std 802.1AXdz-202y.
58       The following reference statement identifies each referenced IEEE

```

⁵*Copyright release for YANG modules:* Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

⁶An ASCII version of the YANG module(s) can be obtained by Web browser from the IEEE 802.1 Website at <https://1.ieee802.org/yang-modules/>.

⁷References in this standard's YANG module definitions are not clickable, as each module has been incorporated unchanged after development and verification using YANG tools.

```
1      Standard as updated by applicable amendments.";
2  reference
3      "IEEE Std 802.1AX-2020 Link Aggregation:
4      IEEE Std 802.1AXdz-202y.";
5  }
6
7  feature cscd {
8      description
9          "Conversation Sensitive Collection and Distribution (CSCD)
10         is supported.";
11      reference
12          "5.3.2, 6.6 of IEEE Std 802.1AX";
13  }
14
15  feature dwc {
16      description
17          "The Discard Wrong Conversation option in CSCD is
18          supported.";
19      reference
20          "5.3.2, 6.6 of IEEE Std 802.1AX";
21  }
22
23  feature sid-map {
24      description
25          "The Service ID to Conversation ID map is supported for
26          use by distribution algorithms.";
27  }
28
29  feature c-vid-dist-alg {
30      description
31          "Supports distribution based on C-VIDs.";
32  }
33
34  feature s-vid-dist-alg {
35      description
36          "Supports distribution based on S-VIDs.";
37  }
38
39  feature i-sid-dist-alg {
40      description
41          "Supports distribution based on I-SIDs.";
42  }
43
44  feature te-sid-dist-alg {
45      description
46          "Supports distribution based on TE-SIDs.";
47  }
48
49  feature flow-hash-dist-alg {
50      description
51          "Supports distribution based on Flow Hash.";
52  }
53
54  feature basic-link-maps {
55      description
56          "Pre-defined admin-conv-link-map configurations to distribute
57          packets on either one or two active links, with any
58          remaining links available as standby.";
59  }
60
61  identity distribution-algorithm {
62      description
63          "Each distribution algorithm is identified by a sequence of
64          4 octets, structured as shown in Figure 8-1. Distribution
65          algorithm identifiers are used by network administrators to
66          select between algorithms and, in Conversation-sensitive
67          LACP and (if supported) Distributed Resilient Network
68          Interconnect (DRNI) operation, to check whether partners
69          and neighbors are using the same algorithm.
70
71          This identity is intended to serve as base identity, not
72          to be directly referenced.
```

```
1      Vendor specific, combination (ex: multi-layer), and other
2      customized distribution algorithms can be created as
3      their own identities in their own YANG files, derived from
4      this imported base type. When the identity description
5      specifies a 4 octet value, this value is used for cscd
6      purposes, and included in LACPDUs. Otherwise the identity
7      is used purely for local selection of a distribution
8      algorithm, and the unspecified distribution algorithm is
9      used for cscd purposes.";
10     reference
11     "8.1, 8.2 of IEEE Std 802.1AX";
12 }
13
14 identity unspecified {
15     base distribution-algorithm;
16     description
17     "The 'Unspecified distribution algorithm' identifier has
18     been reserved for use when the algorithm is unknown (or
19     is not advertised).
20     The algorithm identifier is 00-80-C2-00.";
21     reference
22     "Table 8-1 of IEEE Std 802.1AX";
23 }
24
25 identity c-vids-nomap {
26     if-feature "c-vid-dist-alg";
27     base distribution-algorithm;
28     description
29     "Distribution based on C-VIDs (8.2.1). No Service ID
30     mapping table is used.
31     The algorithm identifier is 00-80-C2-01.";
32     reference
33     "Table 8-1 of IEEE Std 802.1AX";
34 }
35
36 identity c-vids-map {
37     if-feature "c-vid-dist-alg and sid-map";
38     base distribution-algorithm;
39     description
40     "Distribution based on C-VIDs (8.2.1). A Service ID
41     mapping table is used.
42     The algorithm identifier is 00-80-C2-81.";
43     reference
44     "Table 8-1 of IEEE Std 802.1AX";
45 }
46
47 identity s-vids-nomap {
48     if-feature "s-vid-dist-alg";
49     base distribution-algorithm;
50     description
51     "Distribution based on S-VIDs (8.2.2). No Service ID
52     mapping table is used.
53     The algorithm identifier is 00-80-C2-02.";
54     reference
55     "Table 8-1 of IEEE Std 802.1AX";
56 }
57
58 identity s-vids-map {
59     if-feature "s-vid-dist-alg and sid-map";
60     base distribution-algorithm;
61     description
62     "Distribution based on S-VIDs (8.2.2). A Service ID
63     mapping table is used.
64     The algorithm identifier is 00-80-C2-82.";
65     reference
66     "Table 8-1 of IEEE Std 802.1AX";
67 }
68
69 identity i-sids-nomap {
70     if-feature "i-sid-dist-alg";
71     base distribution-algorithm;
```

```
1  description
2    "Distribution based on I-SIDs (8.2.3). No Service ID
3    mapping table is used.
4    The algorithm identifier is 00-80-C2-03.";
5  reference
6    "Table 8-1 of IEEE Std 802.1AX";
7  }
8
9  identity i-sids-map {
10   if-feature "i-sid-dist-alg and sid-map";
11   base distribution-algorithm;
12   description
13     "Distribution based on I-SIDs (8.2.3). A Service ID
14     mapping table is used.
15     The algorithm identifier is 00-80-C2-83.";
16   reference
17     "Table 8-1 of IEEE Std 802.1AX";
18 }
19
20 identity te-sids-nomap {
21   if-feature "te-sid-dist-alg";
22   base distribution-algorithm;
23   description
24     "Distribution based on TE-SIDs (8.2.4). No Service ID
25     mapping table is used.
26     The algorithm identifier is 00-80-C2-04.";
27   reference
28     "Table 8-1 of IEEE Std 802.1AX";
29 }
30
31 identity te-sids-map {
32   if-feature "te-sid-dist-alg and sid-map";
33   base distribution-algorithm;
34   description
35     "Distribution based on TE-SIDs (8.2.4). A Service ID
36     mapping table is used.
37     The algorithm identifier is 00-80-C2-84.";
38   reference
39     "Table 8-1 of IEEE Std 802.1AX";
40 }
41
42 identity flow-hash-nomap {
43   if-feature "flow-hash-dist-alg";
44   base distribution-algorithm;
45   description
46     "Distribution based on Flow Hash (8.2.5). No Service ID
47     mapping table is used.
48     The algorithm identifier is 00-80-C2-05.";
49   reference
50     "Table 8-1 of IEEE Std 802.1AX";
51 }
52
53 identity flow-hash-map {
54   if-feature "flow-hash-dist-alg and sid-map";
55   base distribution-algorithm;
56   description
57     "Distribution based on Flow Hash (8.2.5). A Service ID
58     mapping table is used.
59     The algorithm identifier is 00-80-C2-85.";
60   reference
61     "Table 8-1 of IEEE Std 802.1AX";
62 }
63
64 identity link-map-patterns {
65   description
66     "Base identity for patterns filling the admin-conv-link-map.
67     This identity is intended to serve as base identity, not
68     to be directly referenced. Use of the identity allows vendors
69     to augment the module with vendor specified patterns,";
70   reference
71     "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
72 }
```

```
1 identity one-plus-n {
2   if-feature "basic-link-maps";
3   base link-map-patterns;
4   description
5     "Provides active/standby behavior with one link carrying
6     all traffic and any other links in the LAG are standby.
7     All packets are mapped to the lowest link number that is
8     active in the aggregation. This is updated whenever a link
9     becomes active or inactive in the aggregation. The pattern
10    supports any number of links with link numbers between 1
11    and 4095. Any links in the aggregation with a link number
12    greater than 4095 are not used.
13    The map consists of an identical entry for each CID, with
14    a list of numbers from 1 to 4095, followed by the number 0.
15    The MD5 digest calculated for this map is
16    0xd2dc4cf241740239408945bd87391286 (hex) or
17    0txM8kF0AjlAiUW9hzkShg== (base64).";
18 }
19
20 identity two-plus-n {
21   if-feature "basic-link-maps";
22   base link-map-patterns;
23   description
24     "Provides basic load-balancing on two links and any other
25     links in the aggregation are standby. All packets with an
26     even CID value are mapped to the link with the lowest link
27     number that is active in the aggregation, and all packets
28     with an odd CID value are mapped to the link with the
29     highest link number that is active in the aggregation.
30     This is updated whenever a link becomes active or inactive
31     in the aggregation. The pattern supports any number of
32     links with link numbers between 1 and 4095. Any links in
33     the aggregation with a link number greater than 4095 are
34     not used.
35     The map consists of an identical entry for each even CID
36     containing a list of numbers from 1 to 4095, followed by
37     the number 0, and an identical entry for each odd CID
38     containing a list of numbers from 4095 to 1, followed by
39     the number 0.
40     The MD5 digest calculated for this map is
41     0xfe8fec2effe6cd1cbd78c2a968aa5d20 (hex) or
42     /o/sLv/mzRy9eMKpaKpdIA== (base64).";
43 }
44
45 identity service-map-patterns {
46   description
47     "Base identity for patterns filling the admin-conv-service-map.
48     This identity is intended to serve as base identity, not
49     to be directly referenced. Use of the identity allows vendors
50     to augment the module with vendor specified patterns,";
51   reference
52     "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
53 }
54
55
56 typedef lacp-state {
57   type bits {
58     bit lacp-activity {
59       position 1;
60       description
61         "Provides administrative control over when LACPDUs are
62         transmitted. A value of '1' indicates Active mode where
63         LACPDUs are sent regardless of partner's lacp-activity
64         value. A value of '0' indicates Passive mode where
65         LACPDUs are sent only when the partner's lacp-activity
66         value is '1' (partner is in Active mode).";
67     }
68     bit lacp-timeout {
69       position 2;
70       description
71         "Provides administrative control over the frequency of
72         received LACPDUs. A value of '1' indicates Short Timeout
```

```
1         (so partner uses frequent transmission). A value of '0'
2         indicates Long Timeout (so partner can use infrequent
3         transmission).";
4     }
5     bit aggregation {
6         position 3;
7         description
8             "Provides administrative control over whether this
9             Aggregation Port can be in a LAG with more than one
10            member. A value of '1' indicates the port can be
11            aggregated with other ports. A value of '0' indicates
12            the port can only be a solitary link.";
13    }
14    bit synchronization {
15        position 4;
16        description
17            "The Synchronization state of the MUX state machine.";
18    }
19    bit collecting {
20        position 5;
21        description
22            "The Collecting state of the MUX state machine.";
23    }
24    bit distributing {
25        position 6;
26        description
27            "The Distributing state of the MUX state machine.";
28    }
29    bit defaulted {
30        position 7;
31        description
32            "Indicates the port is using the partner-admin values
33            to select an Aggregator.";
34    }
35    bit expired {
36        position 8;
37        description
38            "The Expired state of the Receive state machine.";
39    }
40 }
41 description
42     "LACP state values as transmitted in LACPDUs.
43     Administrative control over the values of lacp-activity,
44     lacp-timeout, aggregation, and (in partner-admin-lACP-state)
45     synchronization is allowed.
46     The remaining bits are read-only.";
47 reference
48     "6.4.1, 6.4.2.3 of IEEE Std 802.1AX";
49 }
50
51 typedef bridge-reserved-addresses {
52     type enumeration {
53         enum nearest-customer-bridge {
54             description
55                 "Reserved MAC address 01-C2-80-00-00";
56         }
57         enum slow-protocols-multicast {
58             description
59                 "Reserved MAC address 01-C2-80-00-02";
60         }
61         enum nearest-non-tpmr-bridge {
62             description
63                 "Reserved MAC address 01-C2-80-00-03";
64         }
65         enum nearest-bridge {
66             description
67                 "Reserved MAC address 01-C2-80-00-0E";
68         }
69     }
70     description
71         "Group addresses that are filtered by some or all types of
72         bridges in order to limit the scope of propagation of PDUs
```

```

1      in a Bridged Network.";
2      reference
3      "8.6.3 of IEEE Std 802.1Q";
4  }
5
6  grouping link-map {
7      description
8      "Specifies the contents of the admin-conv-link-map.";
9      reference
10     "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
11     choice method {
12         default "cid-list";
13         description
14         "Provides two ways to specify the map contents.";
15         leaf pattern {
16             type identityref {
17                 base link-map-patterns;
18             }
19             description
20             "Use a predefined pattern to fill the map.";
21         }
22         list cid-list {
23             key "cid";
24             description
25             "Data structure to map a Conversation Identifier
26             (CID) to a Link Number. Each entry consists of a CID
27             and a list of link numbers that can potentially be
28             selected for that CID. An empty list of link-numbers
29             means that no links are selected for the CID.";
30             leaf cid {
31                 type uint16 {
32                     range "0..4095";
33                 }
34                 description
35                 "Port Conversation Identifier";
36             }
37             leaf-list link-numbers {
38                 type uint16;
39                 description
40                 "List of zero or more Link Numbers that can
41                 potentially be selected for distribution of frames
42                 with this CID.";
43             }
44         }
45     }
46 }
47
48 grouping service-map {
49     description
50     "Specifies the contents of the admin-conv-service-map.";
51     reference
52     "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
53     choice method {
54         default "cid-list";
55         description
56         "Provides two ways to specify the map contents.";
57         leaf pattern {
58             type identityref {
59                 base service-map-patterns;
60             }
61             description
62             "Use a predefined pattern to fill the map.";
63         }
64         list cid-list {
65             key "cid";
66             description
67             "Data structure to map service identifiers to
68             conversation identifiers. Each entry consists of a
69             Conversation ID (CID) and a list of zero or more
70             Service Identifiers (SIDs) that map to it. An empty
71             list of SIDs means there are no SIDs that map to this
72             CID, and results in the same behavior as not having an

```

```
1         entry for this CID.";
2     leaf cid {
3         type uint16 {
4             range "0..4095";
5         }
6         description
7             "Port Conversation Identifier";
8     }
9     leaf-list service-ids {
10        type uint32;
11        description
12            "List of zero or more SIDs that map to the CID.";
13    }
14 }
15 }
16 }
17 }
18
19
```

20 10.7.2 The *ieee802-dot1ax-linkagg* YANG module

```
21 module ieee802-dot1ax-linkagg {
22     yang-version 1.1;
23     namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-linkagg";
24     prefix dot1ax;
25
26     import ieee802-dot1ax-types {
27         prefix ax;
28     }
29     import ieee802-types {
30         prefix ieee;
31     }
32     import ietf-yang-types {
33         prefix yang;
34     }
35     import ietf-interfaces {
36         prefix if;
37     }
38     import iana-if-type {
39         prefix ianaif;
40     }
41
42     organization
43         "IEEE 802.1 Working Group";
44     contact
45         "WG-URL: http://www.ieee802.org/1/
46         WG-Email: stds-802-1-1@ieee.org
47
48         Contact: IEEE 802.1 Working Group Chair
49         Postal: C/O IEEE 802.1 Working Group
50                IEEE Standards Association
51                45 Hoes Lane
52                Piscataway, NJ 08854
53                USA
54         E-mail: stds-802-1-chairs@ieee.org";
55     description
56         "This YANG module describes the configuration model for Link
57         Aggregation, as specified in IEEE Std 802.1AX, including Link
58         Aggregation Control Protocol (LACP) and Conversation Sensitive
59         Collection and Distribution.
60
61         Copyright (C) IEEE (2025).
62
63         This version of this YANG module is part of IEEE Std 802.1AX;
64         see the standard itself for full legal notices.";
65
66     revision 2025-05-22 {
67         description
68             "Published as part of IEEE Std 802.1AXdz-202y.
69             The following reference statement identifies each referenced IEEE
```

```

1      Standard as updated by applicable amendments.";
2  reference
3      "IEEE Std 802.1AX-2020 Link Aggregation:
4      IEEE Std 802.1AXdz-202y.";
5  }
6
7  typedef key-group-ref {
8      type leafref {
9          path "/dotlax:linkagg/dotlax:key-group/dotlax:name";
10     }
11     description
12         "This type is used by aggregators and aggregation ports to
13         reference an entry in the key-group list.";
14 }
15
16 typedef agg-system-ref {
17     type leafref {
18         path "/dotlax:linkagg/dotlax:agg-system/dotlax:name";
19     }
20     description
21         "This type is used in the key-group list entries to
22         reference an entry in the agg-system list.";
23 }
24
25 augment "/if:interfaces/if:interface" {
26     when "derived-from-or-self(if:type,'ianaif:ieee8023adLag')";
27     description
28         "Applies to interfaces representing a LAG.";
29 }
30 description
31     "Augment Interface with Aggregator parameters.";
32 container lag {
33     description
34         "Contains the configuration and status information that
35         allows an instance of an Aggregator to be managed.";
36     leaf key-group-name {
37         type key-group-ref;
38         mandatory true;
39         description
40             "Specifies the entry in the link-aggregation key-group
41             list to which this aggregator is assigned.";
42     }
43     leaf mac-address {
44         type ieee:mac-address;
45         config false;
46         description
47             "The MAC address assigned to the Aggregator.";
48         reference
49             "7.3.1.1.9 of IEEE Std 802.1AX";
50     }
51     container lacp {
52         config false;
53         description
54             "Contains aggregator LACP operational data.";
55         leaf actor-oper-key {
56             type uint16;
57             description
58                 "The current operational value of the Key for the
59                 Aggregator. The meaning of particular Key
60                 values is of local significance.";
61             reference
62                 "7.3.1.1.8 of IEEE Std 802.1AX";
63         }
64         leaf partner-system {
65             type ieee:mac-address;
66             description
67                 "The value of the MAC address portion of the
68                 System Identifier for the current
69                 protocol partner of this Aggregator. A value
70                 of zero indicates that there is no known partner.
71                 If the aggregation is manually configured, this
72                 System identifier value is assigned by the

```

```
1         local System.";
2     reference
3         "7.3.1.1.10 of IEEE Std 802.1AX";
4 }
5 leaf partner-system-priority {
6     type uint16;
7     description
8         "Indicates the priority value portion of the current
9         Partner's System Identifier. If the aggregation is
10        manually configured, this System Priority value is
11        assigned by the local System.";
12    reference
13        "7.3.1.1.11 of IEEE Std 802.1AX";
14 }
15 leaf partner-oper-key {
16     type uint16;
17     description
18         "The current operational value of the Key for the
19         Aggregators current protocol Partner. If the
20         aggregation is manually configured, this Key value
21         is assigned by the local System.";
22    reference
23        "7.3.1.1.12 of IEEE Std 802.1AX";
24 }
25 leaf aggregate-or-individual {
26     type boolean;
27     description
28         "Indicates whether the Aggregator represents an
29         Aggregate (TRUE) or an Individual link (FALSE).";
30    reference
31        "7.3.1.1.6 of IEEE Std 802.1AX";
32 }
33 }
34 container cscd {
35     if-feature "ax:cscd";
36     config false;
37     description
38         "Aggregator parameters obtained by the operation of LACP
39         supporting CSCD.";
40     leaf partner-port-algorithm {
41         type identityref {
42             base ax:distribution-algorithm;
43         }
44         description
45             "Operational value of the distribution algorithm in
46             use by the LACP Partner.";
47         reference
48             "7.3.1.1.40 of IEEE Std 802.1AX";
49     }
50     leaf partner-conv-service-digest {
51         type binary;
52         description
53             "The MD5 Digest of the admin-conv-service-map in use
54             by the LACP Partner.";
55         reference
56             "7.3.1.1.42, 6.6.3.1 of IEEE Std 802.1AX";
57     }
58     leaf partner-conv-link-digest {
59         type binary;
60         description
61             "The MD5 Digest of the admin-conv-link-map in use
62             by the LACP Partner.";
63         reference
64             "7.3.1.1.41, 6.6.3.1 of IEEE Std 802.1AX";
65     }
66     leaf discard-wrong-conversation {
67         type boolean;
68         description
69             "The operational value that determines whether an
70             Aggregator discards a frame that is collected
71             from an Aggregation Port that is different from the
72             Aggregation Port to which the Aggregator would
```

```

1         distribute a frame with the same Port Conversation
2         ID.";
3     reference
4         "7.3.1.1.37, 6.6 of IEEE Std 802.1AX";
5     }
6     leaf-list active-lag-links {
7         type uint16;
8         description
9             "A list, possibly empty, of the operational
10             link-number of each Aggregation Port active
11             (i.e. Collecting) on this Aggregator.";
12         reference
13             "7.3.1.1.43 of IEEE Std 802.1AX";
14     }
15 }
16 }
17 }
18
19 augment "/if:interfaces/if:interface/if:statistics" {
20     when '../dot1ax:lag' {
21         description
22             "Applies to aggregators.";
23     }
24     description
25         "Augment interface statistics with aggregator statistics.";
26     container lag-stats {
27         config false;
28         description
29             "Contains the set of stats associated with the
30             Aggregator.";
31         leaf unknown-protocol-frames {
32             type yang:counter64;
33             description
34                 "A count of data frames discarded on reception by all
35                 ports that are (or have been) members of the
36                 aggregation, due to the detection of an unknown Slow
37                 Protocols PDU (7.3.3.1.5)";
38             reference
39                 "7.3.1.1.29 of IEEE Std 802.1AX";
40         }
41     }
42 }
43
44 augment "/if:interfaces/if:interface" {
45     description
46         "Augment interface model with Aggregation port
47         configuration nodes.";
48     container aggport {
49         presence
50             "When present, this interface supports Link Aggregation";
51         description
52             "Contains Aggregation Port configuration related nodes,
53             which provides the basic management controls necessary
54             to allow an instance of an Aggregation Port to be managed,
55             for the purposes of Link Aggregation.";
56         leaf key-group-name {
57             type key-group-ref;
58             mandatory true;
59             description
60                 "Specifies the entry in the link-aggregation key-group
61                 list to which this aggregation-port is assigned.";
62         }
63         leaf actor-port-number {
64             type uint16 {
65                 range "1..65535";
66             }
67             description
68                 "The port number assigned to the Aggregation Port.
69                 The port number is communicated in LACPDUs as the
70                 Actor_Port.
71                 This leaf provides the ability to administratively
72                 override the initial value provided by the system.";

```

```
1     reference
2     "7.3.2.1.14, 6.4.6 of IEEE Std 802.1AX";
3 }
4 leaf actor-port-priority {
5     type uint16;
6     default "0x8000";
7     description
8     "The priority value assigned to this Aggregation Port.";
9     reference
10    "7.3.2.1.15, 6.4.6 of IEEE Std 802.1AX";
11 }
12 leaf partner-admin-key {
13     type uint16 {
14         range "1..65535";
15     }
16     description
17     "The current administrative value of the Key for the
18     protocol Partner. The assigned value is used, along
19     with the value of port-partner-admin-system-priority,
20     partner-admin-system, partner-admin-port, and
21     partner-admin-port-priority, in order to achieve
22     manually configured aggregation.
23     The default value is the initial value of actor-port-
24     number provided by the system.";
25     reference
26     "7.3.2.1.10 of IEEE Std 802.1AX";
27 }
28 leaf partner-admin-port {
29     type uint16 {
30         range "1..65535";
31     }
32     description
33     "The current administrative value of the port number for
34     the protocol Partner. The assigned value is used, along
35     with the value of partner-admin-system-priority,
36     partner-admin-system, port-partner-admin-key, and
37     partner-admin-port-priority, in order to achieve
38     manually configured aggregation.
39     The default value is the initial value of actor-port-
40     number provided by the system.";
41     reference
42     "7.3.2.1.16 of IEEE Std 802.1AX";
43 }
44 leaf partner-admin-port-priority {
45     type uint16;
46     default "0";
47     description
48     "The current administrative value of the port priority
49     for the protocol Partner. The assigned value is used,
50     along with the value of partner-admin-system-priority,
51     partner-admin-system, partner-admin-key, and
52     partner-admin-port, in order to achieve manually
53     configured aggregation.";
54     reference
55     "7.3.2.1.18 of IEEE Std 802.1AX";
56 }
57 leaf partner-admin-state {
58     type ax:lacp-state {
59         bit lacp-activity {
60             position 1;
61             description
62             "When the LACP_Activity state is set to '0', the
63             transmission of LACPDUs is controlled by the
64             actor-admin-state.LACP_Activity.";
65         }
66         bit lacp-timeout {
67             position 2;
68             description
69             "When the LACP_Timeout is set to '0', LACPDUs
70             are transmitted at the Slow_Periodic_Time.";
71         }
72         bit aggregation {
```

```
1      position 3;
2      description
3          "When the Aggregation state is set to '0', this
4          port cannot be aggregated with any other port.";
5      }
6      bit synchronization {
7          position 4;
8          description
9              "When the Synchronization state is set to '0', this
10             port cannot become active.";
11      }
12  }
13  default "synchronization";
14  description
15      "Provides administrative control over the partner's
16      LACP_Activity, LACP_Timeout, Aggregation, and
17      Synchronization state when the partner's information is
18      unknown (i.e. no LACPDUs are received from the partner).";
19  reference
20      "7.3.2.1.22, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
21  }
22  leaf wtr-time {
23      type uint16 {
24          range "0..32767";
25      }
26      default "0";
27      description
28          "The wait-to-restore (WTR) period, in seconds, that
29          needs to elapse between an Aggregation Port on a LAG
30          coming up (Port_Operational becoming TRUE) and being
31          permitted to become active (transmitting and
32          receiving frames) on the LAG.";
33      reference
34          "7.3.2.1.30 of IEEE Std 802.1AX";
35  }
36  leaf wtr-revertive {
37      type boolean;
38      default "true";
39      description
40          "Controls revertive or non-revertive mode of operation.
41          When TRUE, the Aggregation Port can become active as
42          soon as the wait-to-restore timer expires regardless of
43          the state of other links in the LAG.
44          When FALSE, the Aggregation Port cannot become active
45          unless there are no other links that can become active
46          in the LAG. The default value is TRUE.";
47      reference
48          "7.3.2.1.31 of IEEE Std 802.1AX";
49  }
50  container lacp {
51      config false;
52      description
53          "Contains Aggregation port LACP operational related
54          nodes.";
55      leaf actor-lacp-version {
56          type uint8;
57          description
58              "The version number transmitted in LACPDUs on this
59              Aggregation Port";
60          reference
61              "7.3.2.1.33 of IEEE Std 802.1AX";
62      }
63      leaf actor-oper-key {
64          type uint16;
65          description
66              "The current operational value of the Key for the
67              Aggregation Port. The meaning of particular Key values
68              is of local significance.";
69          reference
70              "7.3.2.1.5 of IEEE Std 802.1AX";
71      }
72      leaf actor-oper-state {
```

```
1      type ax:lacp-state;
2      description
3          "The operational value of the Actor_State as
4          transmitted in LACPDUs.";
5      reference
6          "7.3.2.1.21, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
7  }
8  leaf partner-lacp-version {
9      type uint8;
10     description
11         "The version number in the LACPDU most recently
12         received on this Aggregation Port.";
13     reference
14         "7.3.2.1.34 of IEEE Std 802.1AX";
15 }
16 leaf partner-oper-system-priority {
17     type uint16;
18     description
19         "Indicates the operational value of priority associated
20         with the Partners System ID. The value of this
21         attribute can contain the manually configured value
22         carried in partner-admin-system-priority if there is
23         no protocol Partner.";
24     reference
25         "7.3.2.1.7 of IEEE Std 802.1AX";
26 }
27 leaf partner-oper-system {
28     type ieee:mac-address;
29     description
30         "Represents the MAC address portion of the Aggregation
31         Port's current protocol partner's System Identifier.
32         A value of zero indicates that there is no known
33         protocol Partner.
34         The value of this attribute can contain the manually
35         configured value carried in partner-admin-system if
36         there is no protocol Partner.";
37     reference
38         "7.3.2.1.9 of IEEE Std 802.1AX";
39 }
40 leaf partner-oper-key {
41     type uint16;
42     description
43         "The current operational value of the Key for the
44         protocol Partner. The value of this attribute can
45         contain the manually configured value carried in
46         partner-admin-key if there is no protocol Partner.";
47     reference
48         "7.3.2.1.11 of IEEE Std 802.1AX";
49 }
50 leaf partner-oper-port {
51     type uint16;
52     description
53         "The operational port number assigned by the
54         Aggregation Port's protocol Partner. The value of this
55         attribute can contain the administratively configured
56         value carried in partner-admin-port if there is no
57         protocol Partner.";
58     reference
59         "7.3.2.1.17 of IEEE Std 802.1AX";
60 }
61 leaf partner-oper-port-priority {
62     type uint16;
63     description
64         "The operational priority value assigned by the
65         Aggregation Port's protocol Partner. The value of this
66         attribute can contain the administratively configured
67         value carried in partner-admin-port-priority if there
68         is no protocol Partner.";
69     reference
70         "7.3.2.1.19 of IEEE Std 802.1AX";
71 }
72 leaf partner-oper-state {
```

```
1      type ax:lacp-state;
2      description
3          "The operational value of the partner's LACP state
4          derived from received LACPDUs or, when Defaulted is
5          true, from the partner-admin-state.";
6      reference
7          "7.3.2.1.23, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
8  }
9  leaf aggregate-or-individual {
10     type boolean;
11     description
12         "When true indicates the Aggregation Port can join a
13         LAG consisting of multiple Aggregation Ports.
14         When false, indicates that the Aggregation Port can
15         only operate as an Solitary link because the
16         Aggregation bit is false in either
17         actor-oper-port-state or partner-oper-port-state.";
18     reference
19         "7.3.2.1.24 of IEEE Std 802.1AX";
20 }
21 leaf wtr-waiting {
22     type boolean;
23     description
24         "Indicates the Aggregation Port is inhibited from
25         becoming active for an interval (determined by
26         wtr-time) after becoming operational or while
27         non-revertive operation is being enforced by the
28         Selection Logic.";
29     reference
30         "7.3.2.32 of IEEE Std 802.1AX";
31 }
32 }
33 container cscd {
34     if-feature "ax:cscd";
35     description
36         "Aggregation port parameters for support of CSCD.";
37     leaf admin-link-number {
38         type uint16;
39         description
40             "The Link_Number value for the Aggregation Port,
41             that is unique among all Aggregation Ports in the same
42             key group. This leaf provides the ability to
43             administratively override the initial value provided
44             by the system. A value of 0 is allowed, but
45             results in no frames being distributed to this
46             Aggregation Port.
47             The default value is the initial value of actor-port-
48             number provided by the system.";
49         reference
50             "7.3.2.1.27, 6.6.3.2 of IEEE Std 802.1AX";
51     }
52     leaf partner-link-number {
53         type uint16;
54         config false;
55         description
56             "The last received value of the Partner_Link_Number,
57             or zero if the Aggregation Port is using default
58             values for the Partner or the Partner LACP Version
59             is 1.";
60         reference
61             "7.3.2.1.29 of IEEE Std 802.1AX";
62     }
63     leaf link-number {
64         type uint16;
65         config false;
66         description
67             "The operational link number for this Aggregation Port.
68             The value is either the same as the admin-link-number,
69             or the corresponding value of the LACP partner.";
70         reference
71             "7.3.2.1.28 of IEEE Std 802.1AX";
72     }
73 }
```

```
1     }
2   }
3 }
4
5 augment "/if:interfaces/if:interface/if:statistics" {
6   when '../dotlax:aggport' {
7     description
8       "Applies to aggregation ports.";
9   }
10  description
11    "Augment interface statistics with aggport statistics.";
12  container aggport-stats {
13    config false;
14    description
15      "Contains stats associated with the Aggregation Port.";
16    leaf lacp-pdu-rx {
17      type yang:counter64;
18      description
19        "The number of valid LACPDUs received on this
20        Aggregation Port.";
21      reference
22        "7.3.3.1.2 of IEEE Std 802.1AX";
23    }
24    leaf marker-pdu-rx {
25      type yang:counter64;
26      description
27        "The number of valid Marker PDUs received on this
28        Aggregation Port.";
29      reference
30        "7.3.3.1.3 of IEEE Std 802.1AX";
31    }
32    leaf marker-response-pdu-rx {
33      type yang:counter64;
34      description
35        "The number of valid Marker Response PDUs received on
36        this Aggregation Port.";
37      reference
38        "7.3.3.1.4 of IEEE Std 802.1AX";
39    }
40    leaf unknown-rx {
41      type yang:counter64;
42      description
43        "The number of frames received that either:
44        a) Carry the Slow Protocols Ethernet Type value,
45        but contain an unknown PDU, or
46        b) Are addressed to the Slow Protocols group MAC
47        Address, but do not carry the Slow Protocols
48        Ethernet Type.";
49      reference
50        "7.3.3.1.5 of IEEE Std 802.1AX";
51    }
52    leaf illegal-rx {
53      type yang:counter64;
54      description
55        "The number of frames received that carry the Slow
56        Protocols Ethernet Type value, but contain a badly
57        formed PDU or an illegal value of Protocol Subtype.";
58      reference
59        "7.3.3.1.6 of IEEE Std 802.1AX";
60    }
61    leaf lacp-pdu-tx {
62      type yang:counter64;
63      description
64        "The number of LACPDUs transmitted on this
65        Aggregation Port.";
66      reference
67        "7.3.3.1.7 of IEEE Std 802.1AX";
68    }
69    leaf marker-pdu-tx {
70      type yang:counter64;
71      description
72        "The number of Marker PDUs transmitted on this
```

```
1         Aggregation Port.";
2     reference
3         "7.3.3.1.8 of IEEE Std 802.1AX";
4 }
5 leaf marker-response-pdu-tx {
6     type yang:counter64;
7     description
8         "The number of Marker Response PDUs transmitted on
9         this Aggregation Port.";
10    reference
11        "7.3.3.1.9 of IEEE Std 802.1AX";
12 }
13 }
14 }
15
16 container linkagg {
17     description
18         "Link Aggregation System specific configuration nodes.";
19     list agg-system {
20         key "name";
21         description
22             "List of aggregation systems.";
23         leaf name {
24             type string;
25             description
26                 "Name for the aggregation system.";
27         }
28         leaf actor-system {
29             type ieee:mac-address;
30             description
31                 "The part of the System Identifier that is a globally
32                 unique MAC address. This leaf provides the ability to
33                 administratively override the initial value provided
34                 by the system.";
35             reference
36                 "7.3.1.1.4, 7.3.2.1.3 of IEEE Std 802.1AX";
37         }
38         leaf actor-system-priority {
39             type uint16;
40             default "0x8000";
41             description
42                 "The part of the System Identifier that is the
43                 priority of the system.";
44             reference
45                 "7.3.1.1.5, 7.3.2.1.2 of IEEE Std 802.1AX";
46         }
47     }
48     list key-group {
49         key "name";
50         unique "actor-admin-key";
51         description
52             "List of key groups. A key group is the set of aggregators
53             and aggregation ports that share the same system priority,
54             system identifier, and aggregation key, and therefore can
55             potentially form a Link Aggregation Group. Each entry in
56             the key group list contains the parameters common to all
57             aggregation ports and/or aggregators in the key group.";
58         reference
59             "6.3.5, 6.4.12 of IEEE Std 802.1AX";
60         leaf name {
61             type string;
62             description
63                 "Name for the key group.";
64         }
65         leaf actor-admin-key {
66             type uint16 {
67                 range "1..65535";
68             }
69             mandatory true;
70             description
71                 "The administrative value of the key used by the
72                 Aggregators and Aggregation Ports in this key-group.";
```

```
1     reference
2       "7.3.1.1.7, 7.3.2.1.4 of IEEE Std 802.1AX";
3   }
4   leaf agg-system-name {
5     type agg-system-ref;
6     mandatory true;
7     description
8       "Specifies the aggregation system for this key
9       group.";
10  }
11  leaf actor-protocol-da {
12    type ax:bridge-reserved-addresses;
13    must ". = 'nearest-customer-bridge' or
14          . = 'slow-protocols-multicast' or
15          . = 'nearest-non-tpmr-bridge'" {
16      error-message "Invalid protocol address";
17    }
18    default "slow-protocols-multicast";
19    description
20      "A 6-octet read-write MAC Address value specifying the DA
21      to be used when sending Link Aggregation Control and
22      Marker PDUs. Valid addresses are the
23      nearest-customer-bridge, slow-protocols-multicast,
24      and nearest-non-tpmr-bridge group addresses. The
25      default value is the slow-protocols-multicast address.";
26    reference
27      "7.3.2.2.1, 6.2.10.2 of IEEE Std 802.1AX";
28  }
29  leaf collector-max-delay {
30    type uint16;
31    description
32      "Specifies the maximum delay, in tens of microseconds,
33      between receiving a frame from an Aggregator Port, and
34      either delivering the frame to the Aggregator Client or
35      discarding the frame. A value of zero means the delay
36      is less than the minimum increment (< 10us).
37      This leaf provides the ability to administratively
38      override the initial value provided by the system.";
39    reference
40      "7.3.1.1.32, 6.2.3.1.1, B.3 of IEEE Std 802.1AX";
41  }
42  leaf actor-admin-state {
43    type ax:lacp-state {
44      bit lacp-activity {
45        position 1;
46        description
47          "Setting the LACP_Activity state to '0' means that the
48          transmission of LACPDUs is controlled by the value of
49          partner-oper-state.LACP_Activity.";
50      }
51      bit lacp-timeout {
52        position 2;
53        description
54          "Setting the LACP_Timeout to '0' means that actor uses
55          the Long_Timeout value, allowing the partner to transmit
56          LACPDUs at the Slow_Periodic_Time.";
57      }
58      bit aggregation {
59        position 3;
60        description
61          "Setting the Aggregation state to '0' prevents this
62          port from being aggregated with any other ports.";
63      }
64    }
65    default "lacp-activity lacp-timeout aggregation";
66    description
67      "Provides administrative control over the values of the
68      LACP_Activity, LACP_Timeout, and Aggregation state.";
69    reference
70      "7.3.2.1.20, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
71  }
72  leaf partner-admin-system {
```

```
1     type ieee:mac-address;
2     default "00-00-00-00-00-00";
3     description
4         "The administrative value of the MAC address portion of
5         the Partner's System Identifier.
6         The assigned value is used, along with the value of
7         port-partner-admin-system, partner-admin-key,
8         partner-admin-port, and partner-admin-port-priority,
9         to achieve administratively configured Link
10        Aggregation Groups with a partner that does not run
11        LACP.";
12    reference
13        "7.3.2.1.8 of IEEE Std 802.1AX";
14    }
15    leaf partner-admin-system-priority {
16        type uint16;
17        default "0";
18        description
19            "The administrative value of priority portion of the
20            the Partner's System Identifier. The assigned
21            value is used, along with the value of
22            port-partner-admin-system, partner-admin-key,
23            partner-admin-port, and partner-admin-port-priority,
24            to achieve administratively configured Link
25            Aggregation Groups with a partner that does not run
26            LACP.";
27        reference
28            "7.3.2.1.6 of IEEE Std 802.1AX";
29    }
30    leaf port-algorithm {
31        type identityref {
32            base ax:distribution-algorithm;
33        }
34        default "ax:unspecified";
35        description
36            "Identifies the algorithm used by the Aggregator to
37            assign frames to a Port Conversation ID. Default is
38            the value for an unspecified distribution algorithm.
39            When the identity description specifies a 4 octet value,
40            this value will be used for cscd purposes, and included
41            in LACPDUs. Otherwise this leaf is used purely for
42            local selection of a distribution algorithm, and the
43            unspecified distribution algorithm is used for cscd
44            purposes.";
45        reference
46            "7.3.1.1.33, Table 8-1 of IEEE Std 802.1AX";
47    }
48    leaf-list lags {
49        type if:interface-ref;
50        config false;
51        description
52            "A list of the if:name of aggregators assigned to this
53            key group.";
54        reference
55            "linkagg:key-group";
56    }
57    leaf-list aggports {
58        type if:interface-ref;
59        config false;
60        description
61            "A list of the if:name of aggregation ports assigned to
62            this key group.";
63        reference
64            "linkagg:key-group";
65    }
66    container cscd {
67        if-feature "ax:cscd";
68        description
69            "Contains CSCD parameters that need to be consistent for
70            all aggregation ports and aggregators in the key group.";
71        container admin-conv-service-map {
72            if-feature "ax:sid-map";
```

```
1      description
2          "Data structure to map service identifiers to
3          conversation identifiers. Each entry consists of a
4          Conversation ID (CID) and a list of zero or more
5          Service Identifiers (SIDs) that map to it. An empty
6          list of SIDs means there are no SIDs that map to this
7          CID, and results in the same behavior as not having an
8          entry for this CID.";
9      reference
10         "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
11     uses ax:service-map;
12 }
13 leaf admin-conv-service-digest {
14     type binary;
15     config false;
16     description
17         "The MD5 Digest of the admin-conv-service-map. The
18         value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
19         when the distribution algorithm specified by
20         agg-port-algorithm does not use the
21         admin-conv-service-map.";
22     reference
23         "7.3.1.1.39, 6.6.3.1 of IEEE Std 802.1AX";
24 }
25 container admin-conv-link-map {
26     description
27         "Data structure to map a Conversation Identifier
28         (CID) to a Link Number. Each entry consists of a CID
29         and a list of link numbers that can potentially be
30         selected for that CID. An empty list of link-numbers
31         means that no links are selected for the CID.";
32     reference
33         "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
34     uses ax:link-map;
35 }
36 leaf admin-conv-link-digest {
37     type binary;
38     config false;
39     description
40         "The MD5 Digest of the admin-conv-link-map. The
41         value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64)
42         when the distribution algorithm specified by
43         agg-port-algorithm does not use the
44         admin-conv-link-map.";
45     reference
46         "7.3.1.1.38, 6.6.3.1 of IEEE Std 802.1AX";
47 }
48 leaf admin-discard-wrong-conv {
49     if-feature "ax:dwc";
50     type enumeration {
51         enum force-true {
52             value 1;
53             description
54                 "Indicates that an Aggregator discards a
55                 frame that is collected from an Aggregation Port
56                 that is different from the Aggregation Port to
57                 which the Aggregator would distribute a frame
58                 with the same Port Conversation ID.";
59         }
60         enum force-false {
61             value 2;
62             description
63                 "Indicates that an Aggregator does not discard
64                 a frame that is collected from an Aggregation Port
65                 that is different from the Aggregation Port to
66                 which the Aggregator would distribute a frame with
67                 the same Port Conversation ID. This is the behavior
68                 of the Aggregator when DWC is not supported";
69         }
70         enum auto {
71             value 3;
72             description
```

```
1         "Indicates that the Aggregator behaves as
2         if the value was force-true only when the actor
3         and partner agree on the algorithms (other than
4         unspecified) and mapping tables used to map frames
5         to Aggregation Ports, and behaves as if the value
6         was force-false otherwise.";
7     }
8 }
9 default "force-false";
10 description
11     "Indicates whether an Aggregator discards a
12     frame that is collected from an Aggregation Port
13     that is different from the Aggregation Port to which
14     the Aggregator would distribute a frame with the
15     same Port Conversation ID.";
16 reference
17     "7.3.1.1.35, 6.6 of IEEE Std 802.1AX";
18 }
19 }
20 }
21 }
22 }
23
24
```

25 10.7.3 The *ieee802-dot1ax-drni* YANG module

```
26 module ieee802-dot1ax-drni {
27     yang-version 1.1;
28     namespace "urn:ieee:std:802.1AX:yang:ieee802-dot1ax-drni";
29     prefix dot1ax-drni;
30
31     import ieee802-dot1ax-types {
32         prefix ax;
33     }
34     import ieee802-dot1ax-linkagg {
35         prefix dot1ax;
36     }
37     import ieee802-types {
38         prefix ieee;
39     }
40     import ietf-yang-types {
41         prefix yang;
42     }
43     import ietf-interfaces {
44         prefix if;
45     }
46
47     organization
48         "IEEE 802.1 Working Group";
49     contact
50         "WG-URL: http://www.ieee802.org/1/
51         WG-Email: stds-802-1-1@ieee.org
52
53         Contact: IEEE 802.1 Working Group Chair
54         Postal: C/O IEEE 802.1 Working Group
55                 IEEE Standards Association
56                 45 Hoes Lane
57                 Piscataway, NJ 08854
58                 USA
59         E-mail: stds-802-1-chairs@ieee.org";
60     description
61         "This YANG module describes the configuration model for a
62         Distributed Resilient Network Interface (DRNI) as specified
63         in 802.1AX.
64
65         Copyright (C) IEEE (2025).
66
67         This version of this YANG module is part of IEEE Std 802.1AX;
68         see the standard itself for full legal notices.";
69 }
```

```
1 revision 2025-05-22 {
2   description
3     "Published as part of IEEE Std 802.1AXdz-202y.
4     The following reference statement identifies each referenced IEEE
5     Standard as updated by applicable amendments.";
6   reference
7     "IEEE Std 802.1AX-2020 Link Aggregation:
8     IEEE Std 802.1AXdz-202y.";
9 }
10
11 typedef irp-state {
12   type bits {
13     bit reserved-1 {
14       position 1;
15       description
16         "Reserved for future use. It is set to 0 on
17         transmit and ignored on receipt.";
18     }
19     bit reserved-2 {
20       position 2;
21       description
22         "Reserved for future use. It is set to 0 on
23         transmit and ignored on receipt.";
24     }
25     bit short-timeout {
26       position 3;
27       description
28         "The Short_Timeout flag indicates the Timeout control value
29         in use by the DRCP Receive machine on this IRP. Short Timeout
30         is encoded as a 1; Long Timeout is encoded as a 0.";
31     }
32     bit synchronization {
33       position 4;
34       description
35         "When the Sync flag is TRUE (1), the DRCP Receive machine has
36         determined the Neighbor DRNI System has a compatible
37         configuration for forming a DRNI.";
38     }
39     bit irc-data {
40       position 5;
41       description
42         "When the IRC_Data flag is TRUE (1), the transfer of Up
43         and Down frames is permitted on the IRC.";
44     }
45     bit drni {
46       position 6;
47       description
48         "The DRNI flag is TRUE (1) when this DRNI System is paired
49         with another DRNI System and FALSE (0) otherwise.";
50     }
51     bit defaulted {
52       position 7;
53       description
54         "When the Defaulted flag is TRUE (1), the DRCP Receive machine
55         is using default operational Neighbor information.
56         When FALSE (0), the operational Neighbor information
57         in use has been received in a DRCPDU.";
58     }
59     bit expired {
60       position 8;
61       description
62         "When the Expired flag is TRUE (1), the DRCP Receive machine
63         is in the EXPIRED state.";
64     }
65   }
66   description
67     "IRP state values as transmitted in DRCPDUs.
68     Administrative control over the values of Short_Timeout
69     and IRC_Data is allowed. The remaining bits are read-only.";
70   reference
71     "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
72 }
```

```

1
2 grouping drni-mask {
3   description
4     "Specifies the contents of a bit mask indexed by CID.";
5   leaf-list cid-str {
6     type string {
7       pattern '(0|[1-9][0-9]{0,3})'
8         + '((-0|[1-9][0-9]{0,3})'
9         + '(%0|[1-9][0-9]{0,3}))?)?'
10        + '(',
11        + '(0|[1-9][0-9]{0,3})'
12        + '((-0|[1-9][0-9]{0,3})'
13        + '(%0|[1-9][0-9]{0,3}))?)?'
14        + ')*';
15    }
16    default "0-4095";
17    description
18      "A comma separated list of single CID values (x),
19      CID ranges (x-y), or ranges with an increment (x-y%z).
20      For example, the string '0,10-13,100-110%4' sets the
21      bits in the mask corresponding to CIDs
22      0,10,11,12,13,100,104,108.
23      Some special cases are:
24      'x-y%z' where x+z>y is the same as 'x'.
25      'x-y%0' is the same as 'x'.
26      'x-y%1' is the same as 'x-y'.
27      'x-y%2' sets the bit for all even CIDs in the range
28      when x is even, and for all odd CIDs when x is odd.
29      Although the pattern allows values of x, y, and z in the
30      range 0-9999, resulting CID values greater than 4095
31      have no effect on the mask.
32      Specified as a list of strings, so that long strings
33      can be broken into several shorter strings.
34      The default sets all bits in the mask.";
35  }
36  leaf invert-mask {
37    type boolean;
38    default "false";
39    description
40      "When true the specified mask is inverted:
41      each zero replaced with a one,
42      and each one replaced with a zero.";
43  }
44 }
45
46 augment "/if:interfaces/if:interface/dotlax:lag" {
47   description
48     "Augmentation parameters only for Aggregators with
49     DRNI enabled.";
50   container drni {
51     presence "When present, this Aggregator is enabled for DRNI";
52     description
53       "Aggregator parameters to support a Distributed
54       Resilient Network Interface";
55     leaf irp-name {
56       type if:interface-ref;
57       mandatory true;
58       description
59         "Interface Name (if:name) of the Port supporting the
60         Intra Relay Port (IRP) of this DRNI Gateway.";
61       reference
62         "7.4.1.1.23 of IEEE Std 802.1AX";
63     }
64     leaf drni-aggregator-key {
65       type uint16 {
66         range "1..65535";
67       }
68       description
69         "The Aggregator Key value to be used by the Aggregator
70         supporting this DRNI Gateway (and the Aggregation Ports
71         assigned to this DRNI Gateway) when paired with a
72         neighbor DRNI System via the IRC. The default value is

```

```

1         the Aggregator's Actor_Admin_Aggregator_Key value.";
2     reference
3         "7.4.1.1.15 of IEEE Std 802.1AX";
4 }
5 leaf drni-aggregator-system {
6     type ieee:mac-address;
7     default "00-00-00-00-00-00";
8     description
9         "The Aggregator System value to be used by the
10        Aggregator supporting this DRNI Gateway (and the
11        Aggregation Ports assigned to this DRNI Gateway)
12        when paired with a neighbor DRNI System via the
13        Intra-Relay Connection (IRC).";
14     reference
15         "7.4.1.1.13 of IEEE Std 802.1AX";
16 }
17 leaf drni-aggregator-system-priority {
18     type uint16;
19     default "0";
20     description
21         "The Aggregator System Priority value to be used by the
22        Aggregator supporting this DRNI Gateway (and the
23        Aggregation Ports assigned to this DRNI Gateway) when
24        paired with a neighbor DRNI System via the IRC.";
25     reference
26         "7.4.1.1.14 of IEEE Std 802.1AX";
27 }
28 leaf drcp-protocol-da {
29     type ax:bridge-reserved-addresses;
30     must "." = 'nearest-customer-bridge' or
31         . = 'nearest-bridge' or
32         . = 'nearest-non-tpmr-bridge'" {
33         error-message "Invalid protocol address";
34     }
35     default "nearest-non-tpmr-bridge";
36     description
37         "A 6-octet read-write MAC Address value specifying the
38        Destination Address for Distributed Relay Control PDUs
39        transmitted on the Intra-Relay Port. Valid addresses are
40        the nearest-customer-bridge, nearest-bridge, and
41        nearest-non-tpmr-bridge group addresses. The default
42        value is the nearest-non-tpmr-bridge group
43        address.";
44     reference
45         "7.4.1.1.12, 9.6.1.1 of IEEE Std 802.1AX";
46 }
47 leaf home-admin-irp-state {
48     type irp-state {
49         bit short-timeout {
50             position 3;
51             description
52                 "The Short_Timeout flag indicates the Timeout control value
53                 in use by the DRCP Receive machine on this IRP. Short Timeout
54                 is encoded as a 1; Long Timeout is encoded as a 0.";
55         }
56         bit irc-data {
57             position 5;
58             description
59                 "When the IRC_Data flag is TRUE (1), the transfer of Up
60                 and Down frames is permitted on the IRC.";
61         }
62     }
63     default "short-timeout irc-data";
64     description
65         "Provides administrative control over the values of the
66        Short_Timeout and IRC-Data state.";
67     reference
68         "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
69 }
70 leaf home-oper-irp-state {
71     type irp-state;
72     config false;

```

```
1      description
2        "A string of 8 bits, corresponding to the current
3        operational value of IRP_State as transmitted in
4        DRCPDUs.";
5      reference
6        "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
7    }
8    leaf home-cscd-gateway-control {
9      type boolean;
10     default "true";
11     description
12       "When TRUE, allows the DRNI Gateway Port selection to
13       be based on the CSCD parameters that control the
14       Aggregator Port selection.";
15     reference
16       "7.4.1.1.16 of IEEE Std 802.1AX";
17   }
18   leaf home-dr-client-gateway-control {
19     type boolean;
20     default "true";
21     description
22       "When TRUE, allows the Distributed Relay Client to
23       determine whether to forward frames through the DRNI
24       Gateway Port.";
25     reference
26       "7.4.1.1.17 of IEEE Std 802.1AX";
27   }
28   leaf home-gateway-algorithm {
29     type identityref {
30       base ax:distribution-algorithm;
31     }
32     default "ax:unspecified";
33     description
34       "Identifies the algorithm used by the DRNI Gateway to
35       assign frames to a Gateway Conversation ID. 8.2 provides
36       the IEEE 802.1 OUI (00-80-C2) Gateway Algorithm
37       encodings. Default is the value for an unspecified
38       distribution algorithm.";
39     reference
40       "7.4.1.1.6 of IEEE Std 802.1AX";
41   }
42   container home-gateway-conv-service-map {
43     if-feature "ax:sid-map";
44     description
45       "Data structure to map service identifiers to
46       conversation identifiers. Each entry consists of a
47       Conversation ID (CID) and a list of zero or more Service
48       Identifiers (SIDs) that map to it. Frames with Service
49       IDs not contained in the map are not mapped to any
50       Gateway Conversation ID and are discarded.";
51     reference
52       "7.4.1.1.20, 6.6.3.1 of IEEE Std 802.1AX";
53     uses ax:service-map;
54   }
55   leaf home-gateway-conv-service-digest {
56     type binary;
57     config false;
58     description
59       "The MD5 Digest of the home-gateway-conv-service-map. The
60       value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64) when
61       the distribution algorithm specified
62       by agg-port-algorithm does not use the
63       home-gateway-conv-service-map.";
64     reference
65       "7.4.1.1.21 of IEEE Std 802.1AX";
66   }
67   container gateway-enable-mask {
68     description
69       "A vector of Boolean values, with one value for each
70       possible Gateway Conversation ID. A 1 indicates that
71       frames associated with that Gateway Conversation ID
72       are allowed to pass through this Gateway Port, and a
```

```

1         0 indicates that such frames are not allowed to pass.
2         Default value is all bits set to 1.";
3     reference
4         "7.4.1.1.18, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
5     uses drni-mask;
6 }
7 container gateway-preference-mask {
8     description
9         "A vector of Boolean values, with one value for each
10        possible Gateway Conversation ID. A 1 indicates that
11        this Gateway Port is the preferred Gateway when both
12        DRNI Gateways have the Gateway Conversation ID enabled
13        in the gateway-available-mask, and a 0 indicates that
14        it is not preferred.
15        Default value is all bits set to 1.";
16    reference
17        "7.4.1.1.19, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
18    uses drni-mask;
19 }
20 container gateway-available-mask {
21     config false;
22     description
23         "A vector of Boolean values, with one value for each
24        possible Gateway Conversation ID. A 1 indicates that
25        this Gateway Port is eligible to be selected to pass
26        that Gateway Conversation ID, and a 0 indicates that
27        it is not eligible.";
28    reference
29        "7.4.1.1.22, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
30    uses drni-mask;
31 }
32 container neighbor {
33     config false;
34     description
35         "Operational values for the DRNI neighbor obtained
36        from DRCPDUs.";
37     leaf oper-irp-state {
38         type irp-state;
39         description
40             "A string of 8 bits, corresponding to the current
41             operational value of IRP_State as transmitted in
42             DRCPDUs.";
43         reference
44             "7.4.1.1.26, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
45     }
46     leaf system {
47         type ieee:mac-address;
48         description
49             "The MAC Address portion of the System Identifier of
50             the Neighbor DRNI System (connected via the
51             Intra-Relay Port). ";
52         reference
53             "7.4.1.1.29 of IEEE Std 802.1AX";
54     }
55     leaf system-priority {
56         type uint16;
57         description
58             "The priority portion of the System Identifier of the
59             Neighbor DRNI System (connected via the Intra-Relay
60             Port).";
61         reference
62             "7.4.1.1.30 of IEEE Std 802.1AX";
63     }
64     leaf drni-key {
65         type uint16;
66         description
67             "The DRNI key value received from the Neighbor DRNI
68             System (connected via the IntraRelay Port).";
69         reference
70             "7.4.1.1.31 of IEEE Std 802.1AX";
71     }
72     leaf aggregator-algorithm {

```

```
1      type identityref {
2          base ax:distribution-algorithm;
3      }
4      description
5          "The Port algorithm used by the Neighbor Aggregator to
6          assign frames to Port Conversation IDs.";
7      reference
8          "7.4.1.1.33 of IEEE Std 802.1AX";
9  }
10 leaf aggregator-conv-service-digest {
11     type binary;
12     description
13         "The MD5 Digest of the Neighbor Aggregator's
14         Admin_Conv_Service_Map. Obtained from the Home
15         Aggregator State TLV last received from the Neighbor
16         DRNI System.";
17     reference
18         "7.4.1.1.34 of IEEE Std 802.1AX";
19 }
20 leaf aggregator-conv-link-digest {
21     type binary;
22     description
23         "The MD5 Digest of the Neighbor Aggregator's
24         Admin_Conv_Link_Map. Obtained from the Home Aggregator
25         State TLV (9.6.2.4) last received from the Neighbor
26         DRNI System.";
27     reference
28         "7.4.1.1.35 of IEEE Std 802.1AX";
29 }
30 leaf partner-system-priority {
31     type uint16;
32     description
33         "The priority portion of the System Identifier of the
34         Neighbor Aggregator's Partner.";
35     reference
36         "7.4.1.1.36 of IEEE Std 802.1AX";
37 }
38 leaf partner-system {
39     type ieee:mac-address;
40     description
41         "The MAC Address portion of the System Identifier of
42         the Neighbor Aggregator's Partner.";
43     reference
44         "7.4.1.1.37 of IEEE Std 802.1AX";
45 }
46 leaf partner-aggregator-key {
47     type uint16;
48     description
49         "The operational key value of the Neighbor
50         Aggregator's Partner.";
51     reference
52         "7.4.1.1.38 of IEEE Std 802.1AX";
53 }
54 leaf cscd-state {
55     type bits {
56         bit reserved-1 {
57             position 0;
58             description
59                 "Bit 1 is reserved for future use. It is set to 0
60                 and ignored on receipt.";
61         }
62         bit reserved-2 {
63             position 1;
64             description
65                 "Bit 2 is reserved for future use. It is set to 0
66                 and ignored on receipt.";
67         }
68         bit reserved-3 {
69             position 2;
70             description
71                 "Bit 3 is reserved for future use. It is set to 0
72                 and ignored on receipt.";
```

```
1      }
2      bit cscd_gateway_control {
3          position 3;
4          description
5              "CSCD_Gateway_Control is encoded in bit 4. When
6              this flag is TRUE, the DRNI Gateway is configured
7              to minimize forwarding data frames on the IRC by
8              selecting the DRNI Gateway and Aggregator Ports
9              for forwarding any given Conversation ID to be in
10             the same DRNI System.";
11      }
12      bit discard_wrong_conversation {
13          position 4;
14          description
15              "Discard_Wrong_Conversation is encoded in bit 5.
16              The Aggregator's Discard_Wrong_Conversation
17              value.";
18      }
19      bit differ_conv_link_digests {
20          position 5;
21          description
22              "Differ_Conv_Link_Digests is encoded in bit 6.
23              This flag is TRUE when the Aggregator's
24              Actor_Conv_Link_Digest matches the Aggregator's
25              Partner_Conv_Link_Digest.";
26      }
27      bit differ_conv_service_digests {
28          position 6;
29          description
30              "Differ_Conv_Service_Digests is encoded in bit 7.
31              This flag is TRUE when the Aggregator's
32              Actor_Conv_Service_Digest matches the Aggregator's
33              Partner_Conv_Service_Digest.";
34      }
35      bit differ_port_algorithms {
36          position 7;
37          description
38              "Differ_Port_Algorithms is encoded in bit 8. The
39              Aggregator's differPortAlgorithms flag is TRUE
40              when the Aggregator's Actor_Port_Algorithm matches
41              the Aggregator's Partner_Port_Algorithm.";
42      }
43  }
44  description
45      "8 bits, corresponding to the Aggregator_CSCD_State
46      in the Neighbor_Aggregator_State variable. The first
47      three bits (the least significant bits of CSCD_State)
48      are reserved; the fourth bit corresponds to the
49      Neighbor's value for Home_Admin_CSCD_Gateway_Control;
50      the fifth bit corresponds to the Neighbor Aggregator's
51      operational value for Discard_Wrong_Conversation; and
52      the sixth, seventh, and eighth bits correspond to the
53      Neighbor Aggregator's operational value for
54      differConvLinkDigests, differConvServiceDigests, and
55      differPortAlgorithms, respectively, (the most
56      significant bits of CSCD_State).";
57  reference
58      "7.4.1.1.39 of IEEE Std 802.1AX";
59  }
60  leaf-list active-links {
61      type uint16;
62      description
63          "A list of the operational Link_Numbers of Aggregation
64          Ports that are currently active (i.e., collecting) on
65          the Neighbor's Aggregator. An empty list indicates that
66          there are no Aggregation Ports active. Each integer
67          value in the list carries an aAggPortOperLinkNumber
68          attribute value.";
69      reference
70          "7.4.1.1.40 of IEEE Std 802.1AX";
71  }
72  leaf gateway-algorithm {
```

```

1      type identityref {
2          base ax:distribution-algorithm;
3      }
4      description
5          "The gateway algorithm used by the Neighbor DRNI
6          Gateway to assign frames to Gateway Conversation IDs.";
7      reference
8          "7.4.1.1.41 of IEEE Std 802.1AX";
9  }
10 leaf gateway-conv-service-digest {
11     type binary;
12     description
13         "The MD5 Digest of the Neighbor DRNI Gateway's
14         the Home_Admin_Gateway_Conv_Service_Map. Obtained
15         from Gateway_Conv_Service_Digest in the
16         Neighbor_Gateway_State TLV last received from the
17         Neighbor DRNI System.";
18     reference
19         "7.4.1.1.42 of IEEE Std 802.1AX";
20 }
21 container gateway-available-mask {
22     description
23         "A vector of Boolean values, with one value for each
24         possible Gateway Conversation ID. A 1 indicates that
25         the Neighbor DRNI Gateway Port is eligible to be
26         selected to pass that Gateway Conversation ID, and
27         a 0 indicates that it is not eligible.";
28     reference
29         "7.4.1.1.43 of IEEE Std 802.1AX";
30     uses drni-mask;
31 }
32 container gateway-preference-mask {
33     description
34         "A vector of Boolean values, with one value for each
35         possible Gateway Conversation ID. A 1 indicates that
36         the Neighbor DRNI Gateway Port is the preferred
37         Gateway when both DRNI Gateways have the Gateway
38         Conversation ID enabled in the gateway-available-mask,
39         and a 0 indicates that it is not preferred.";
40     reference
41         "7.4.1.1.44 of IEEE Std 802.1AX";
42     uses drni-mask;
43 }
44 }
45 }
46 }
47
48 augment "/if:interfaces/if:interface/if:statistics"
49     + "/dotlax:lag-stats" {
50     when '../dotlax:lag/dotlax-drni:drni' {
51         description
52             "Applies to aggregators with DRNI present.";
53     }
54     description
55         "Augment interface statistics with DRNI statistics.";
56     container drni-stats {
57         description
58             "Contains DRNI specific statistics.";
59         leaf drcpdus-rx {
60             type yang:counter64;
61             description
62                 "The number of valid DRCPDUs received on this
63                 Intra-Relay Port.";
64             reference
65                 "7.4.1.1.45 of IEEE Std 802.1AX";
66         }
67         leaf illegal-rx {
68             type yang:counter64;
69             description
70                 "The number of frames received on this Intra-Relay
71                 Port that carry the DRCP EtherType value,
72                 but contain a badly formed PDU.";

```

```
1      reference
2      "7.4.1.1.46, 9.6.1.4 of IEEE Std 802.1AX";
3  }
4  leaf drcpdus-tx {
5      type yang:counter64;
6      description
7          "The number of valid DRCPDUs transmitted on this
8          Intra-Relay Port.";
9      reference
10     "7.4.1.1.47 of IEEE Std 802.1AX";
11 }
12 }
13 }
14 }
15
16
17
```

¹ Annex A

² (normative)

³ Protocol implementation conformance statement (PICS) ⁴ proforma⁸

⁵ Insert the following rows at the end of the table in A.2.1

A.2.1 Major capabilities/options

Item	Feature	Status	References	Support
YANG	Does the implementation support management operations using YANG modules?	O	10.6	Yes [] No []
YANG modules	Is the ieee802-dot1ax-types module supported?	YANG:M	10.7.1	N/A [] Yes []
	Is the ieee802-dot1ax-linkagg module supported?	YANG:M	10.7.2	N/A [] Yes []
	Is the ieee802-dot1ax-drni module supported?	DRNI AND YANG:M	10.7.3	N/A [] Yes []

⁶

⁸ Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

¹ **Annex G**

² (informative)

³ **Bibliography**

⁴ *Insert the following two bibliography entries after [B8] in Annex G:*

⁵

⁶ [B9] IETF RFC 7803, Changing the Registration Policy for the NETCONF Capability URNs Registry,
⁷ February 2016.

⁸ [B10] IETF RFC 8040, RESTCONF Protocol, January 2017.

⁹ [B11] OMG Unified Modeling Language (OMG UML), Version 2.5, March 2015.⁹

⁹OMG documents are available from the Object Management Group (<https://www.omg.org/>).